



Sailing into the Future of the Semiconductor Industry

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Senior Director

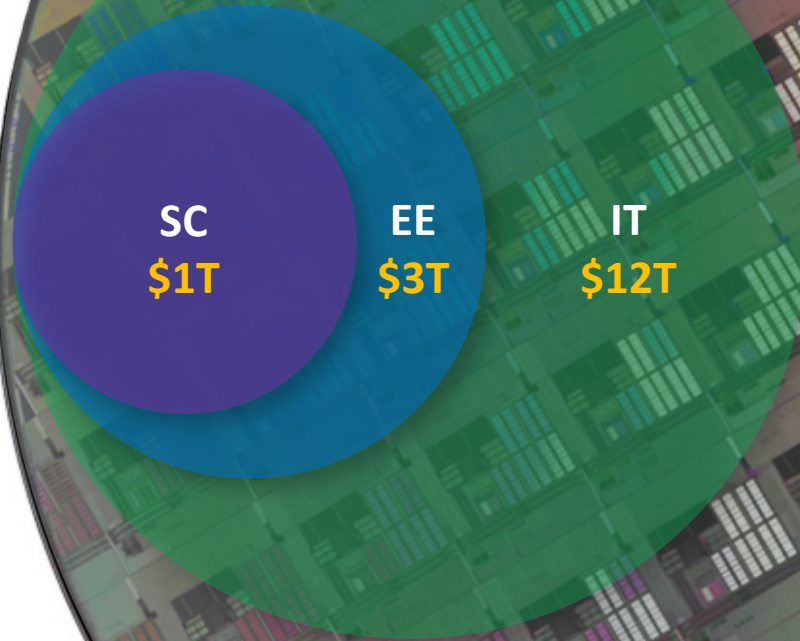
Advanced Technology Business Development, TSMC

Outline

- **Long-term market overview**
- **AI megatrend and impact on**
 - **Data Centers**
 - **Edge devices**
 - **Automotive**
 - **IoT**
- **Advanced Logic**
- **Advanced Packaging**
- **System-Technology Co-Optimization (STCO)**
- **Summary**

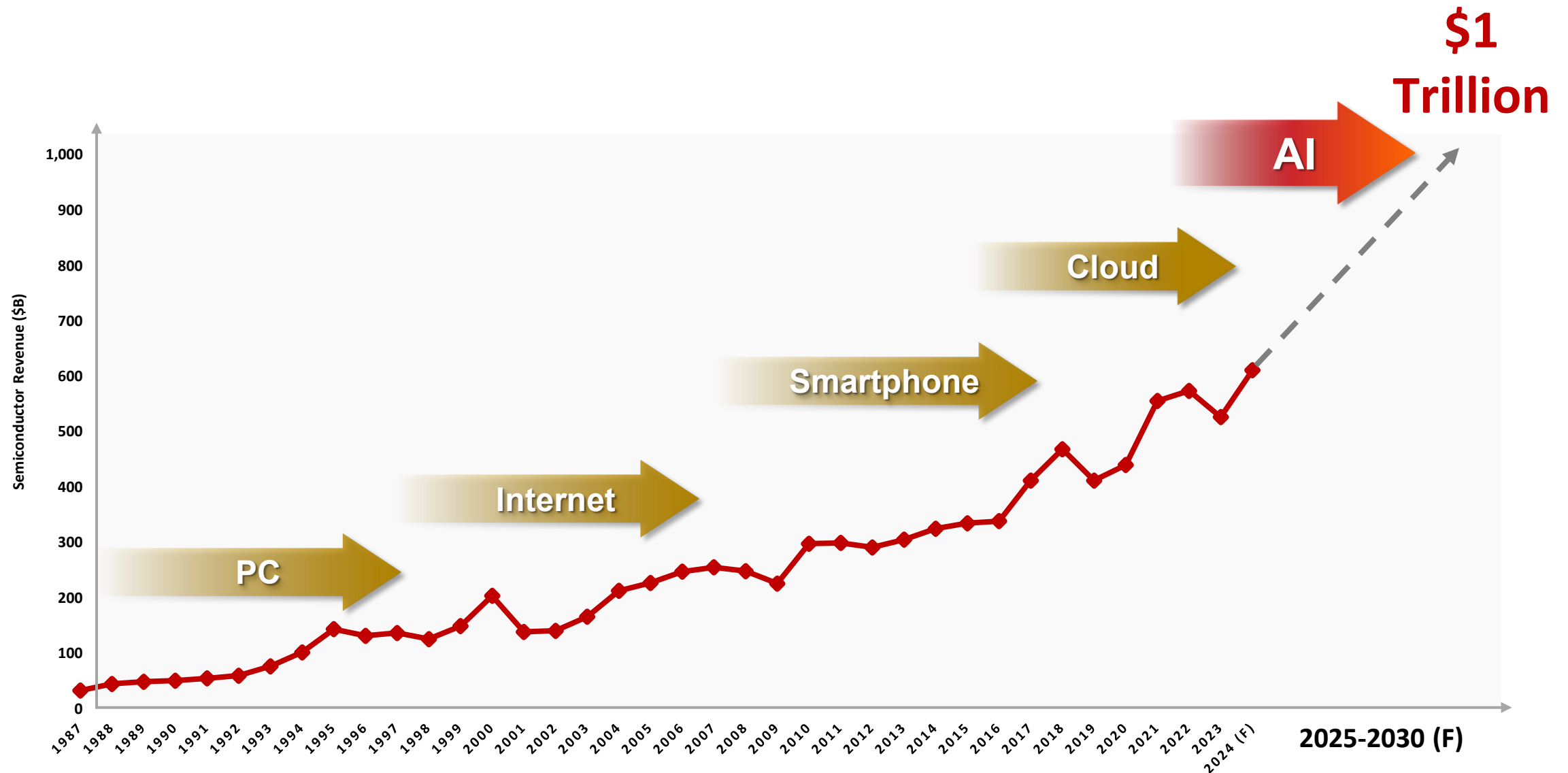
Semiconductor: The Enabler to Power Global Economy

2030



GDP
\$145T

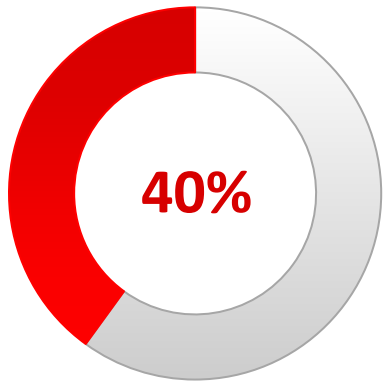
Semiconductor Industry Outlook



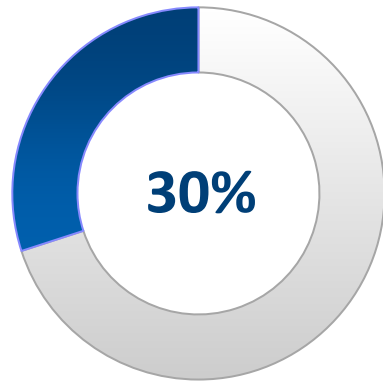
Source: WSTS 2021, TSMC estimates

2023 Semiconductor Market by Platform

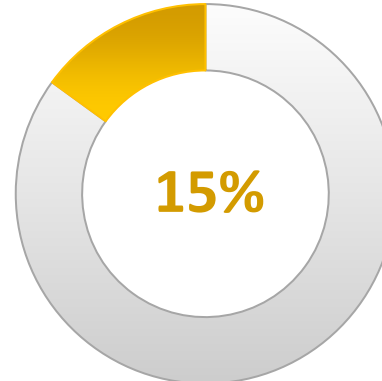
~ \$1 Trillion Market



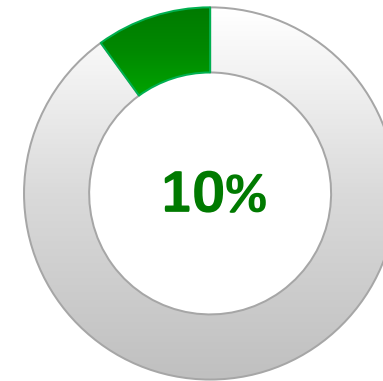
HPC



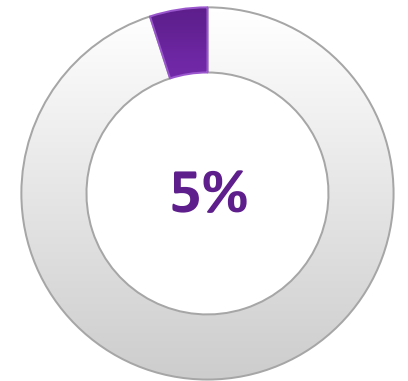
Mobile



Automotive



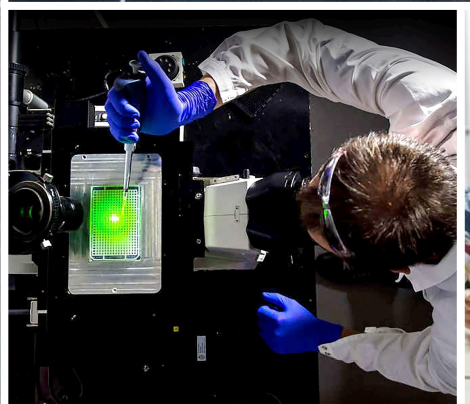
IoT



Others



An AI-infused World is Emerging



AI-empowered Platforms



Datacenter



Edge Device



Automotive



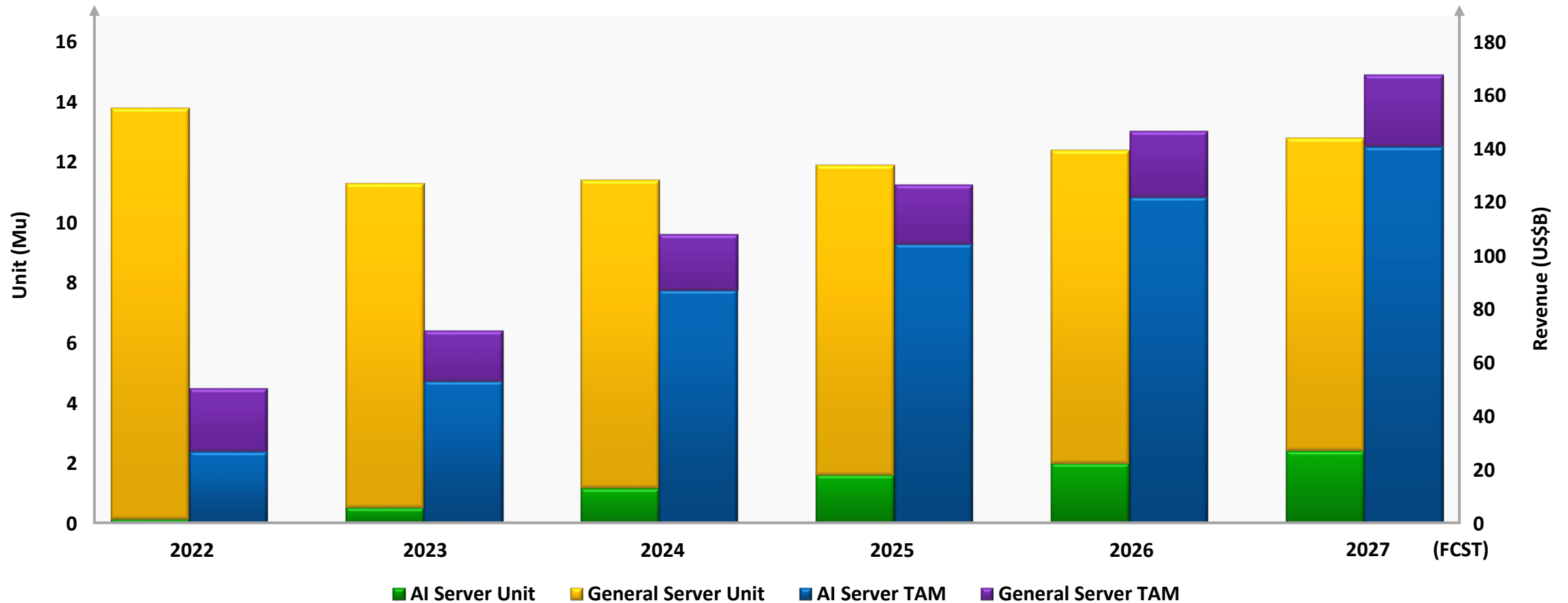
IoT

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AI Server Revenue Forecast

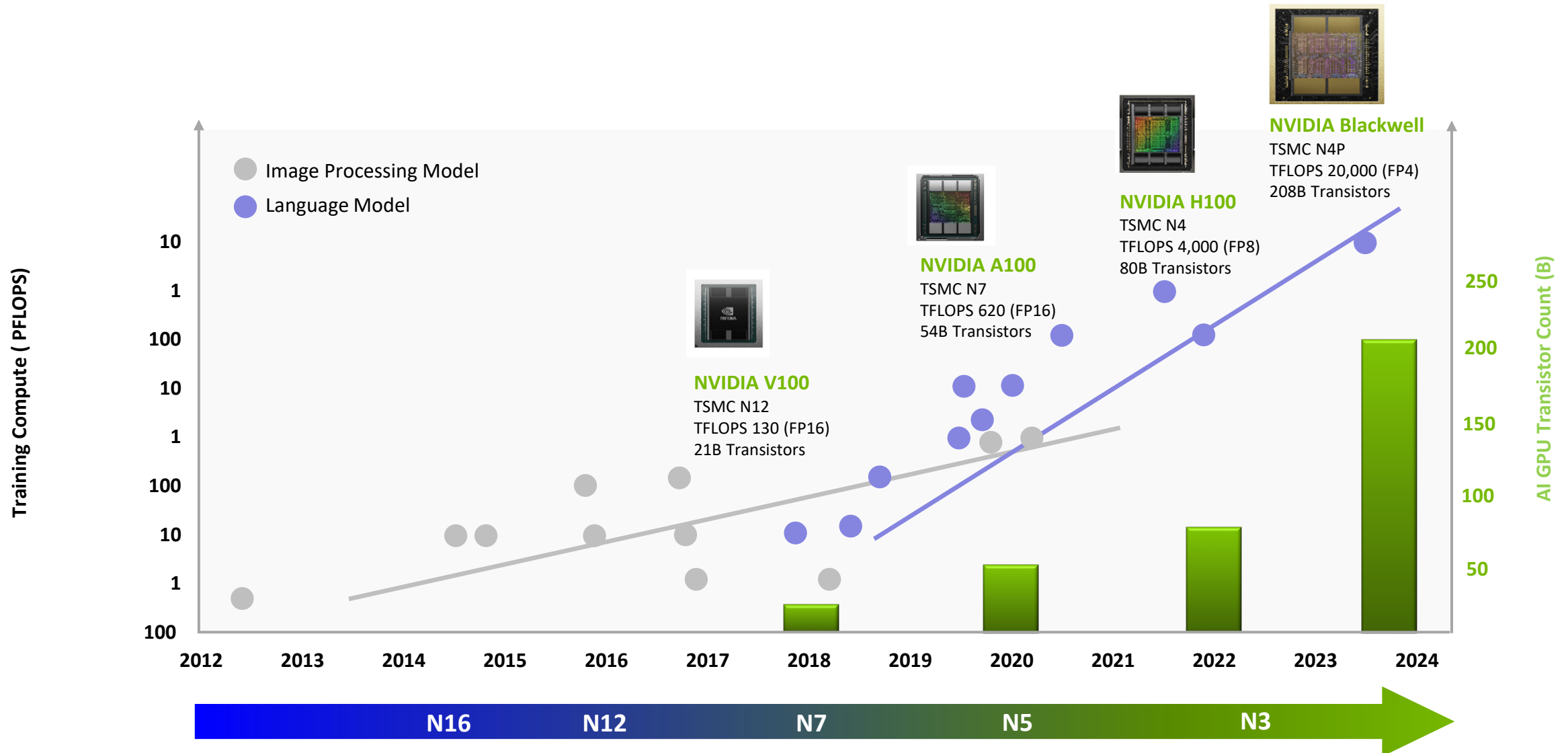
- AI Server ASP is much higher than General Server
- AI Server Unit CAGR is 73%, revenue CAGR 39% from 2022 to 2027



Source: Gartner, Mercury Research and TSMC estimate

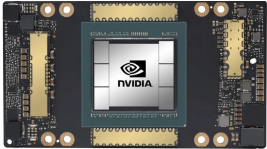
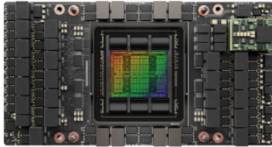
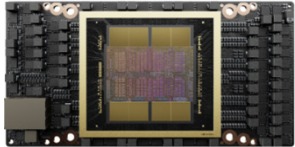
Technology Advancement Ignites Explosive AI Growth

More Compute, Higher Memory Bandwidth, Larger Scale Heterogeneous Integration



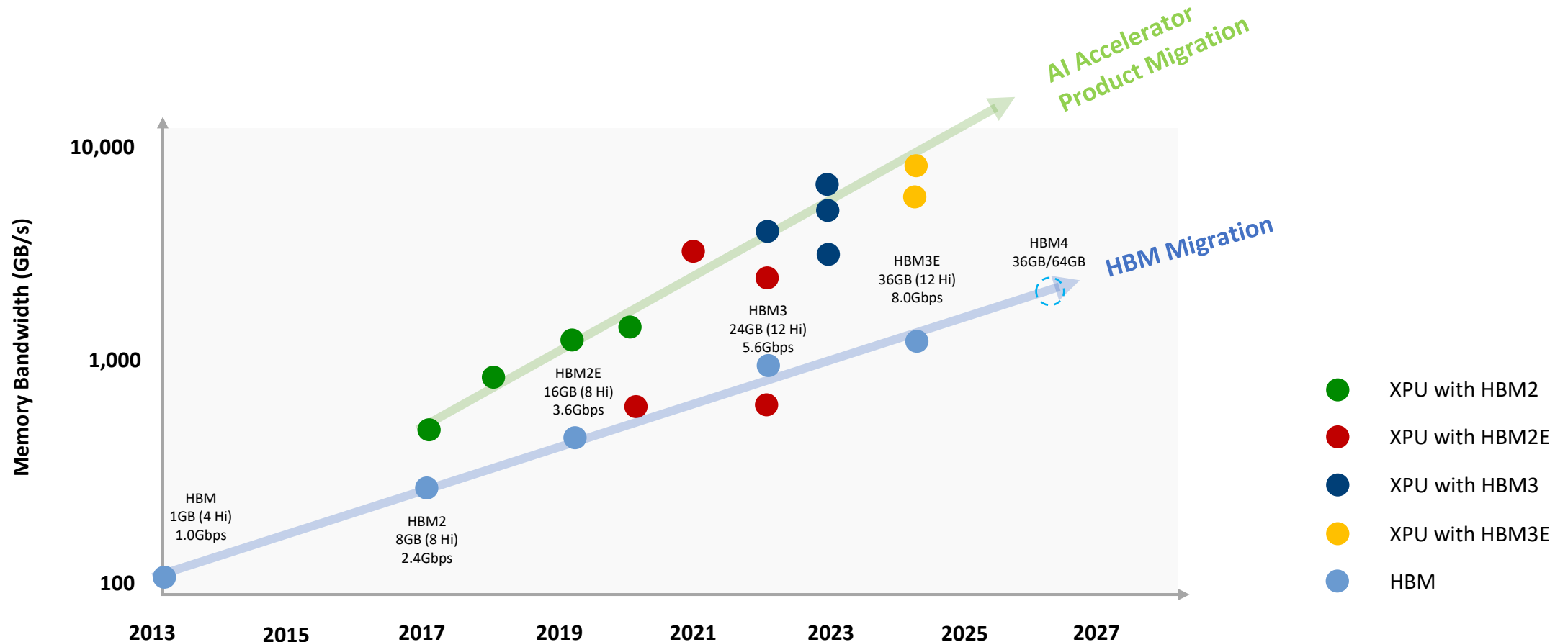
AI Accelerator Migration and Logic/Packaging Technology

- Advanced logic and packaging technology made key contributions to AI accelerators in performance, memory bandwidth and power efficiency

NVIDIA AI Accelerator			
	A100	H200	B200
Process Node	N7	N5	N4P
Die Size (mm2)	826	814	N/A
Gate Count (B)	54	80 (1.48X)	104 (1.3X, x2)
Clock (MHz)	1410	1830 (1.3X)	TBA
BF16 TFLOPS	624	1979 (3.2X)	4500 (2.27X)
Memory Type	HBM2 (x5)	HBM3e (x6)	HBM3e (x8)
Advanced Packaging	CoWoS-S	CoWoS-S	CoWoS-L
Memory BW (GBps)	1555	4800 (3.1X)	8000 (1.7X)
TDP (Watt)	400	700	1000
TFLOPS / Watt	1.56	2.83 (1.8X)	4.5 (1.6X)

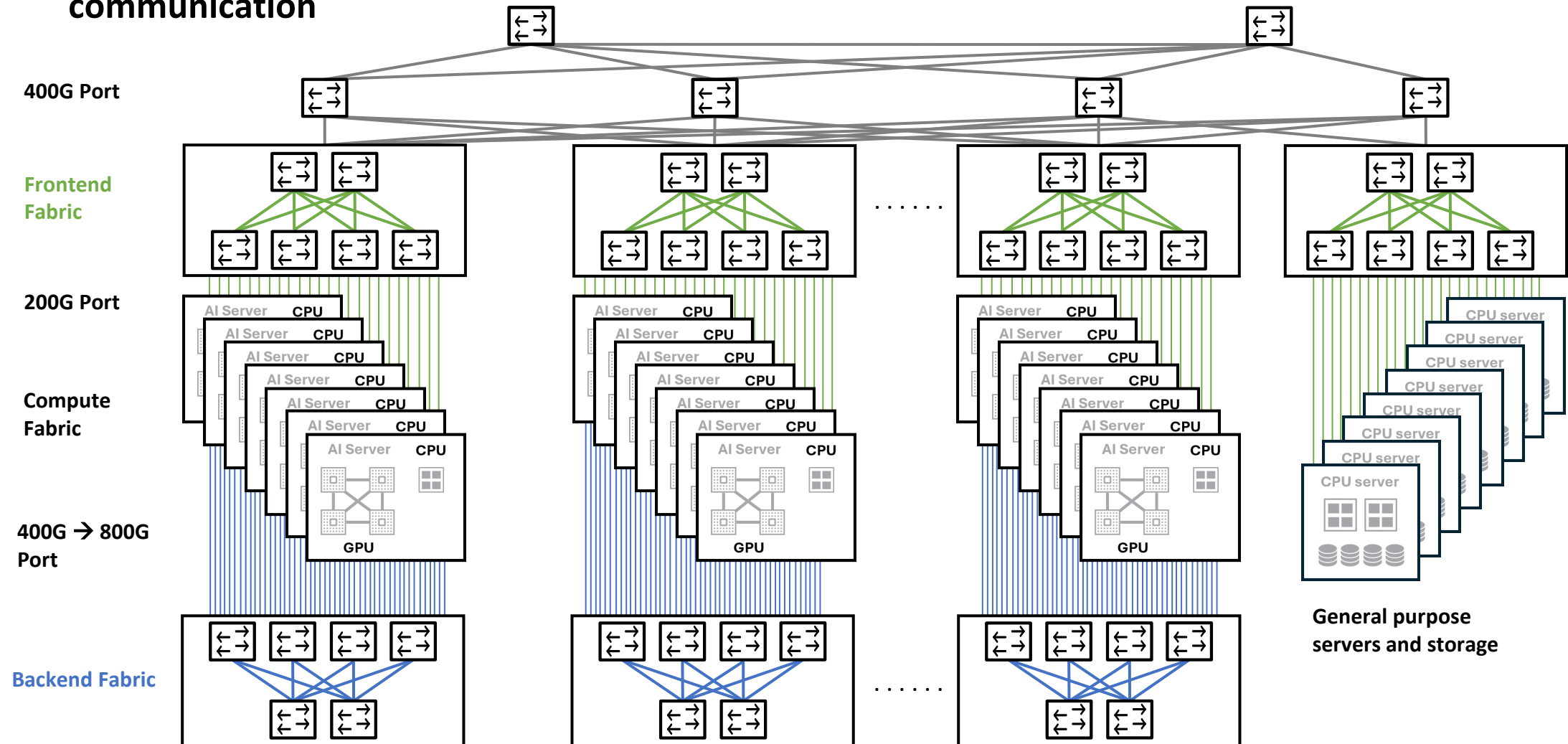
AI Accelerator Memory Bandwidth Outpaces HBM

- AI accelerator instantiates HBM multiple times for memory bandwidth to process LLM workload



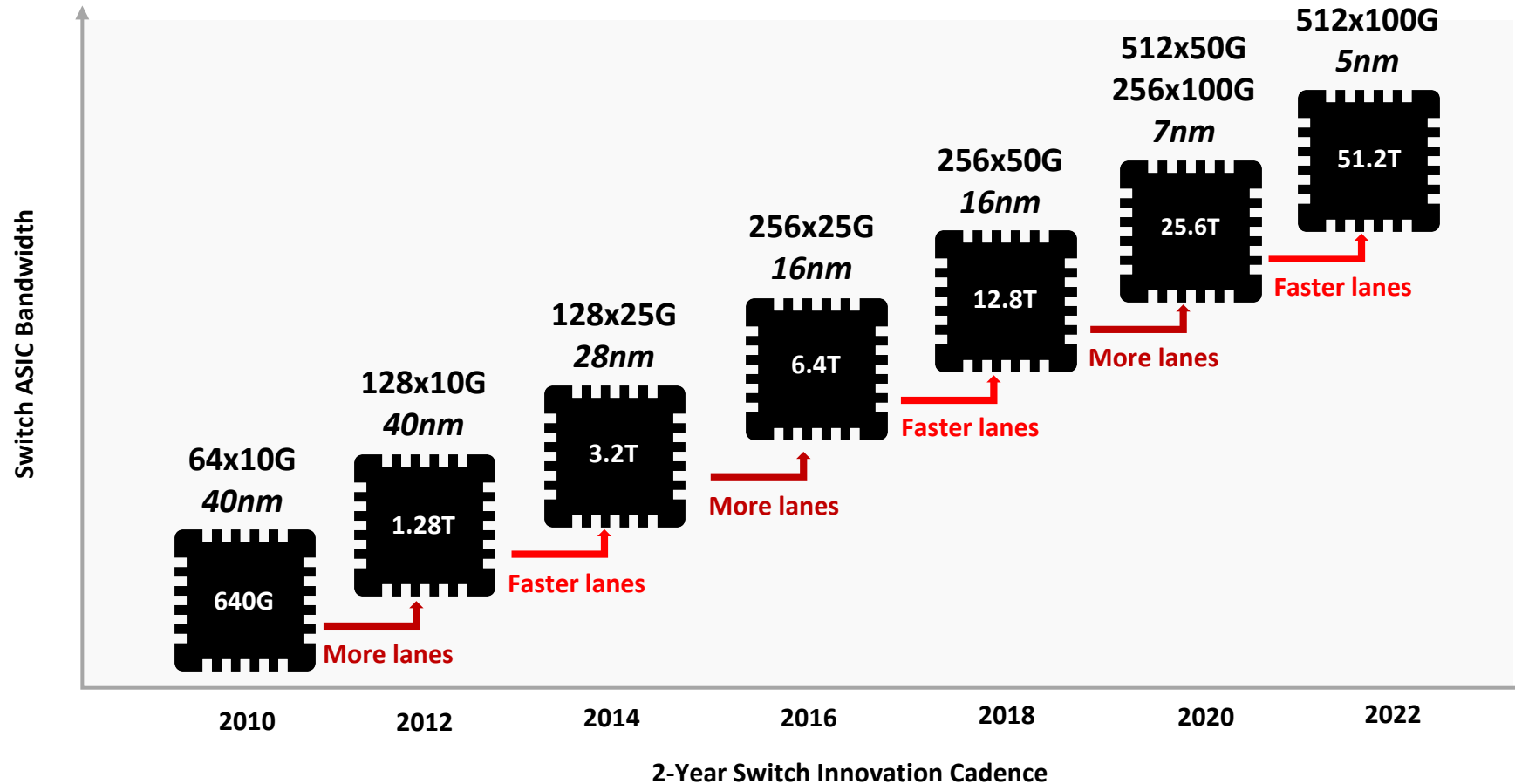
AI Data Center Clusters with Frontend and Backend Networks

- General-purpose server moves data in latency-tolerant north-south traffic to i/f with external users
- AI datacenter requires robust east-west backend data flow to support heavy parallel GPU communication



Datacenter Switch Bandwidth Grows at 2X / 2 Years

- New datacenters require switch solutions at faster data rate and lower power



Source: Broadcom press releases; Cyriel Minkenberg

Switch Product Migration and Logic Technology

- Advanced logic technology made key contributions to switch product in SerDes performance, network bandwidth and power efficiency

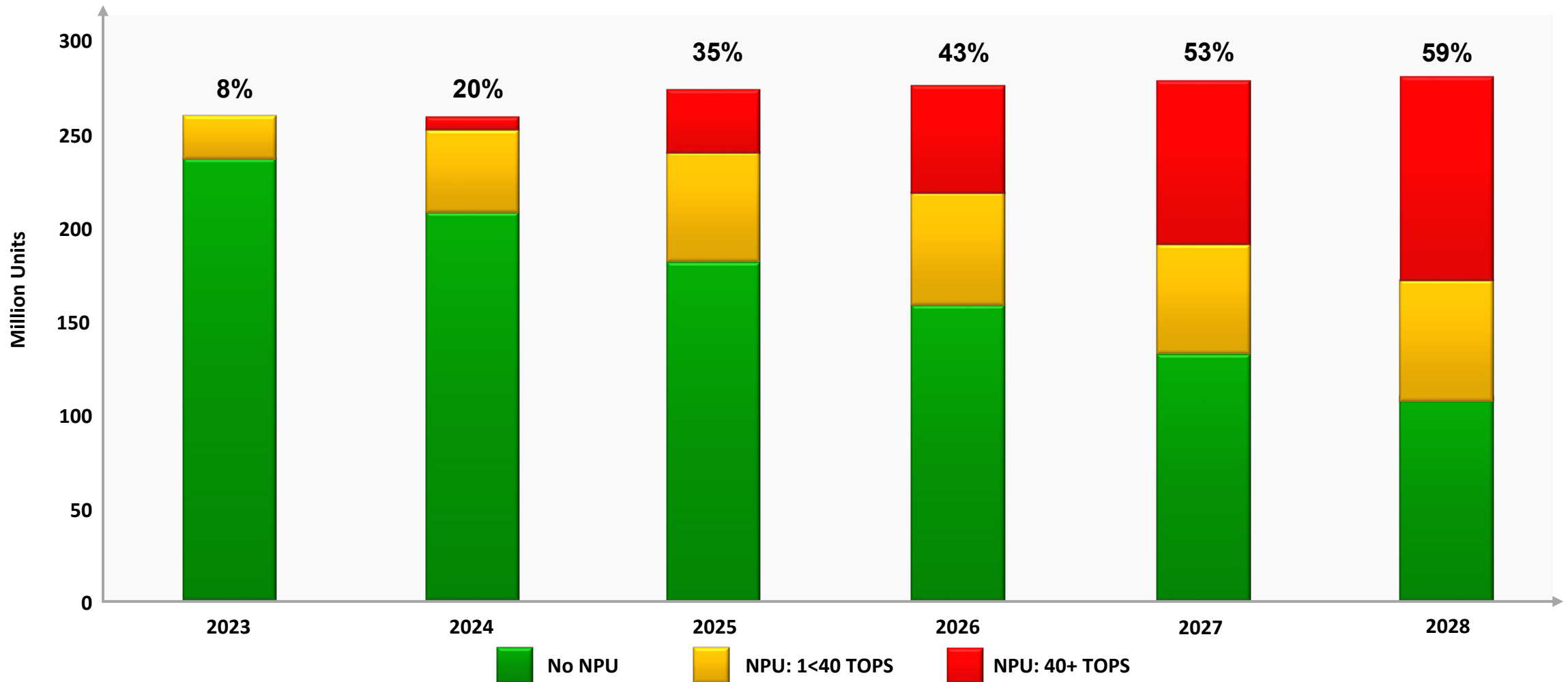
 BROADCOM®	 BCM56980	 BCM56990	 BCM78900
Process Node	N16	N7	N5
Transistor Count	N/A	30B+	60B
SerDes Speed	256 x 53G PAM4	512 x 53G PAM4 (2X)	512 x 106G PAM4 (2X)
Aggregate Bandwidth	12.8 Tb/s	25.6Tb/s (2X)	51.2 Tb/s (2X)
Network Ports	32x 400GbE 64x 200GbE 128x 100GbE	64x 400GbE 128x 200GbE 256x 100GbE	64x 800GbE 128x 400GbE 256x 200GbE
Power Efficiency	1.7 W/100G	1.2 W/100G (0.7X)	0.88 W/100G (0.7X)
TDP	220 W	306 W	450 W
Release Year	2018	2020	2022

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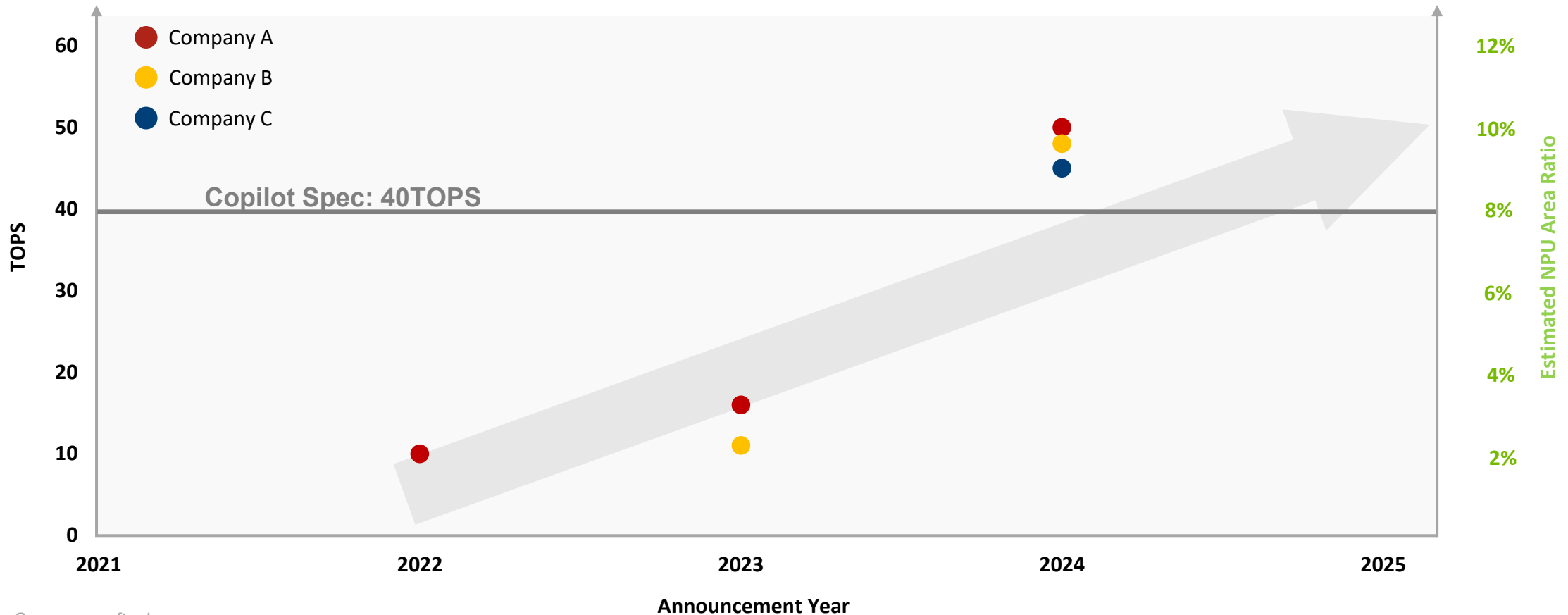
Worldwide AI PC Unit Outlook

- AI PCs to represent more than half of all PC shipments by 2027



Accelerating NPU Performance in AI PC

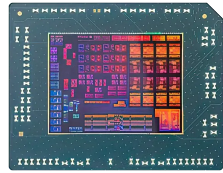
- APUs released in 2022 and 2023 have <40TOPS NPU performance
- APUs offered in 2024 exceed 40TOPS (Microsoft Copilot spec) with increasing NPU area



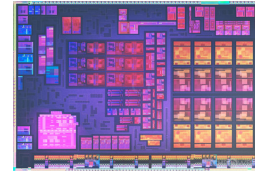
Source: wccftch

PC CPU Migration and TSMC Technology

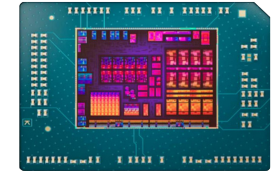
- Advanced logic technology made key contributions to PC CPU in performance and power efficiency



AMD RYZEN 9 6980HS
“REMBRANDT”



AMD RYZEN 9 8945HS
“HAWK POINT”

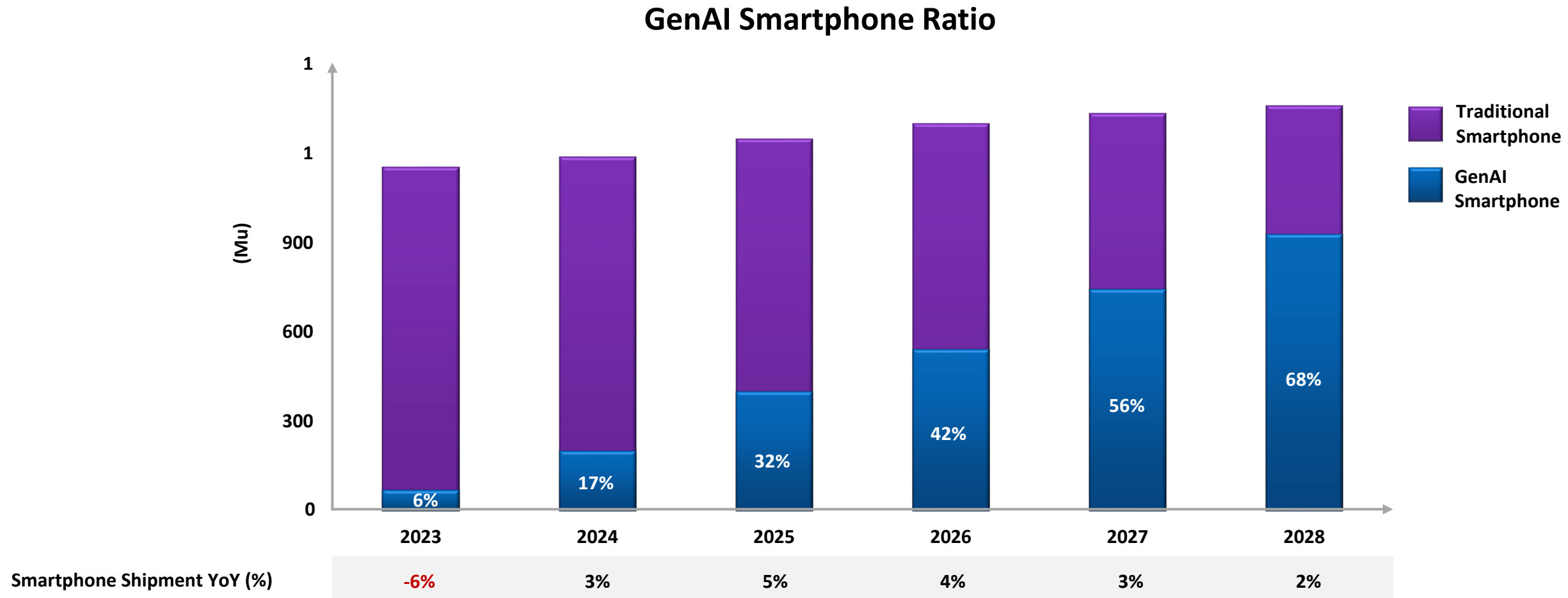


AMD RYZEN AI 9 HX 375
“STRIX POINT”

	AMD RYZEN 9 6980HS “REMBRANDT”	AMD RYZEN 9 8945HS “HAWK POINT”	AMD RYZEN AI 9 HX 375 “STRIX POINT”
Process Node	N6	N4	N4P
CPU Core Architecture	Zen 3+	Zen 4	4 Zen 5 + 8 Zen 5c
Clock Speed (Max)	5.0 GHz	5.2 GHz	5.1 GHz
L2/L3 Cache	4 MB / 16 MB	4 MB / 16 MB	12 MB / 24 MB
GPU Core Architecture	RDNA 2 iGPU (Radeon 680M)	RDNA 3 iGPU (Radeon 780M)	RDNA 3+ iGPU (Radeon 890M)
GPU Cores/Clocks Speed (Max)	12 CUs / 2.4 GHz	12 CUs / 2.8 GHz	16 CUs / 2.9 GHz
NPU Core Architecture	N/A	XDNA	XDNA 2
AI Capability	N/A	16 TOPS	50+ TOPS (3.12X)
Cinebench R23 SC/MC	1,669/14,736	1,810/17,082 (1.08X/1.16X)	2,326/22,246 (1.28X/1.30X)
TDP (W)	15W-55W	35W-54W	15W-54W
Perf/Watt (SC/MC)	30/268	33.5/316 (1.12X/1.18X)	51.7/494 (1.54X/1.56X)
Launch	Q1 2022	Q1 2024	Q3 2024

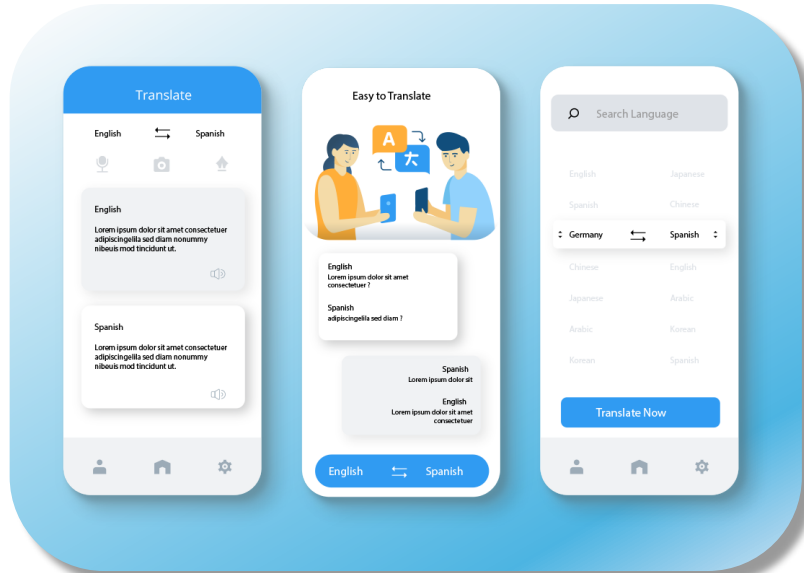
GenAI Smartphone Penetration to Grow 4X from '24 to '28

- GenAI Smartphone: (1) NPU > 38 TOPS, (2) multimodal capabilities, (3) hybrid (cloud & edge), (4) embedded pre-trained SLM, and (5) enhanced user experience
- GenAI Smartphone will re-invigorate high-end segment growth and proliferate into mid-end segment

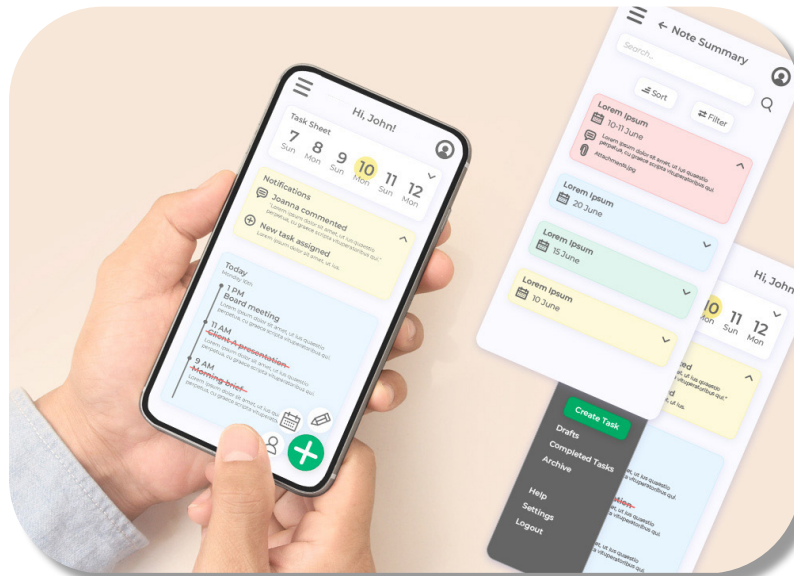


GenAI Smartphone as Personal Assistant

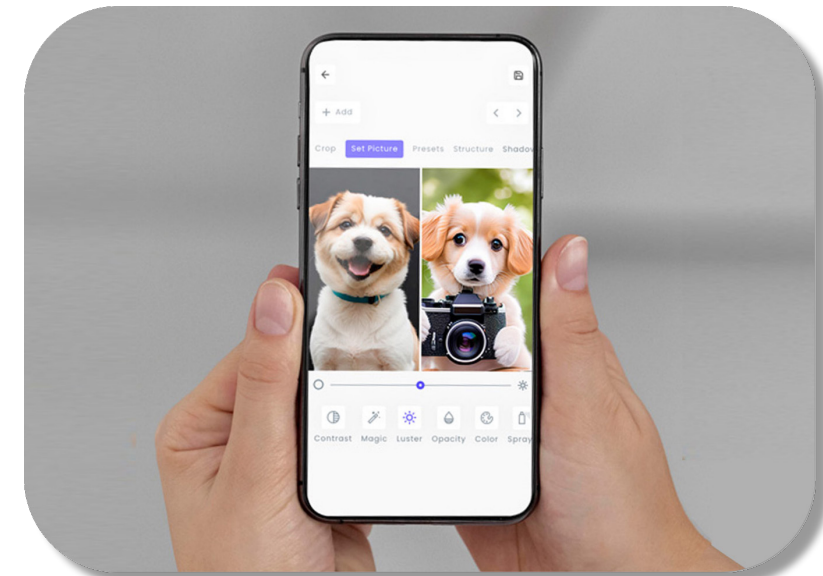
Real-time Call / Text Translation



Note Summary

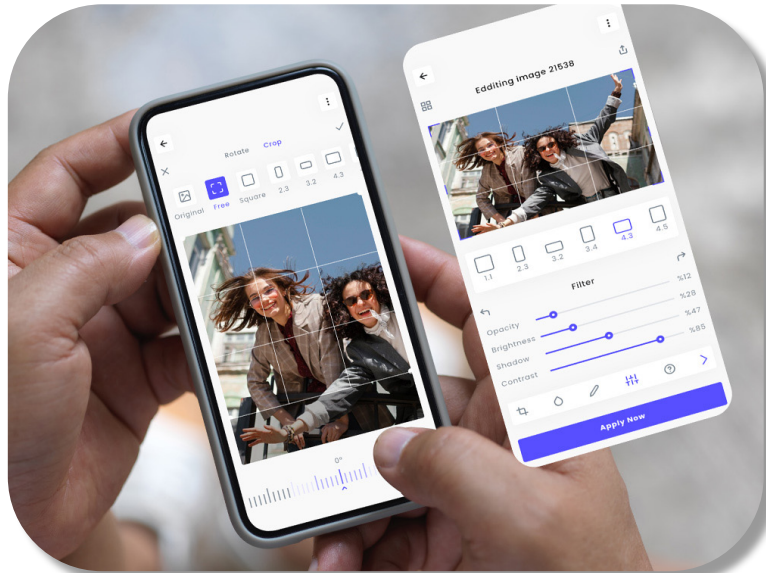


Text / Image / Video Generation

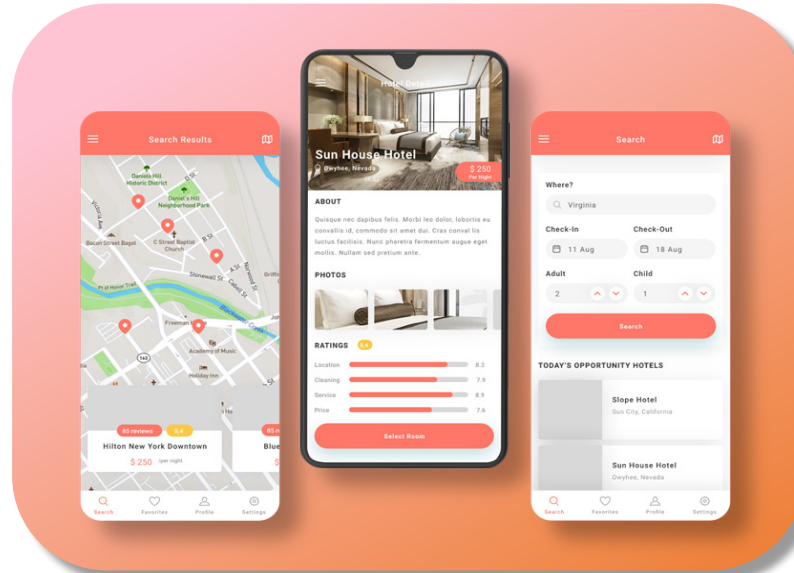


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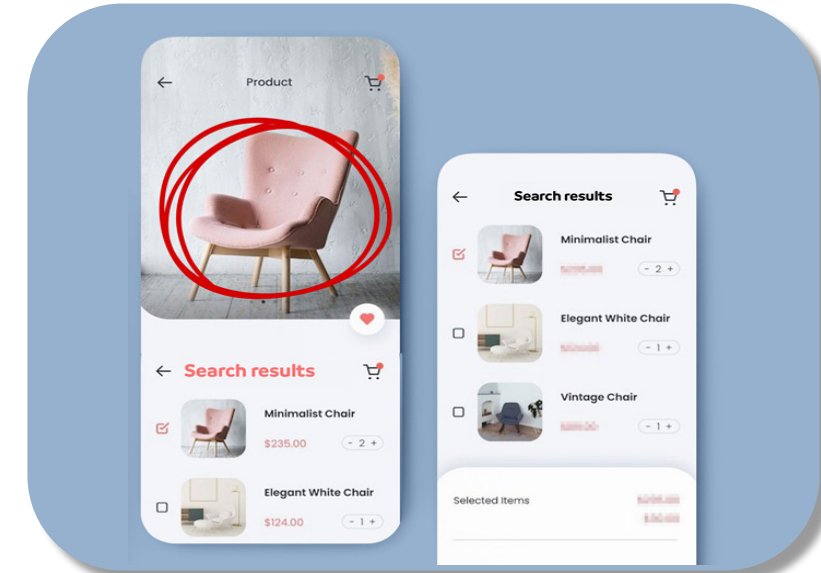
Photo Edit and Creation



Itinerary Arrangement

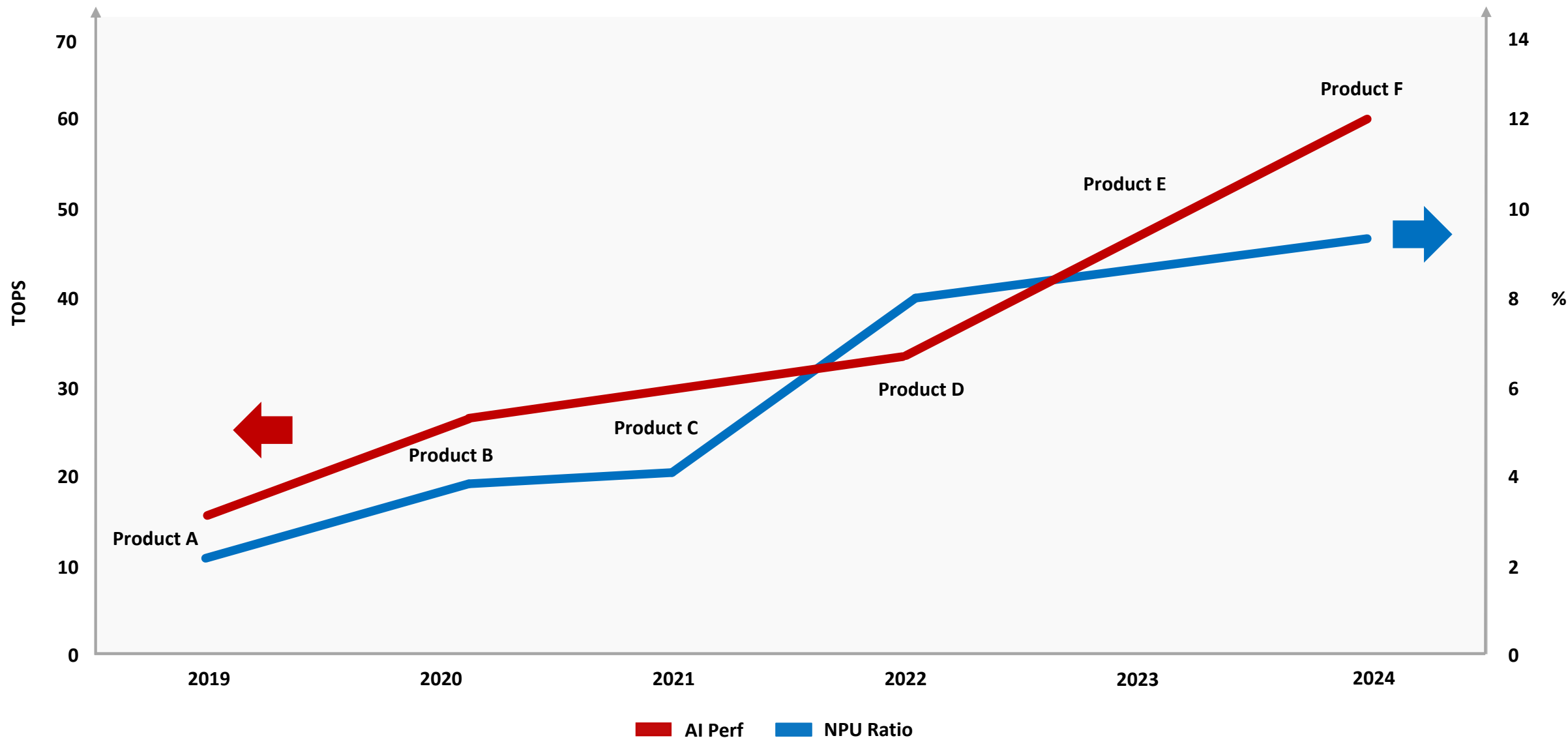


Circle to Search



NPU Content Increases in Premium Smartphone SoC

- NPU area ration increases in premium smartphone SoC to support on-device GenAI



Flagship Mobile SoC Evolution and Logic Technology

- Advanced logic technology make key contributions in CPU / GPU / AI performance under power budget

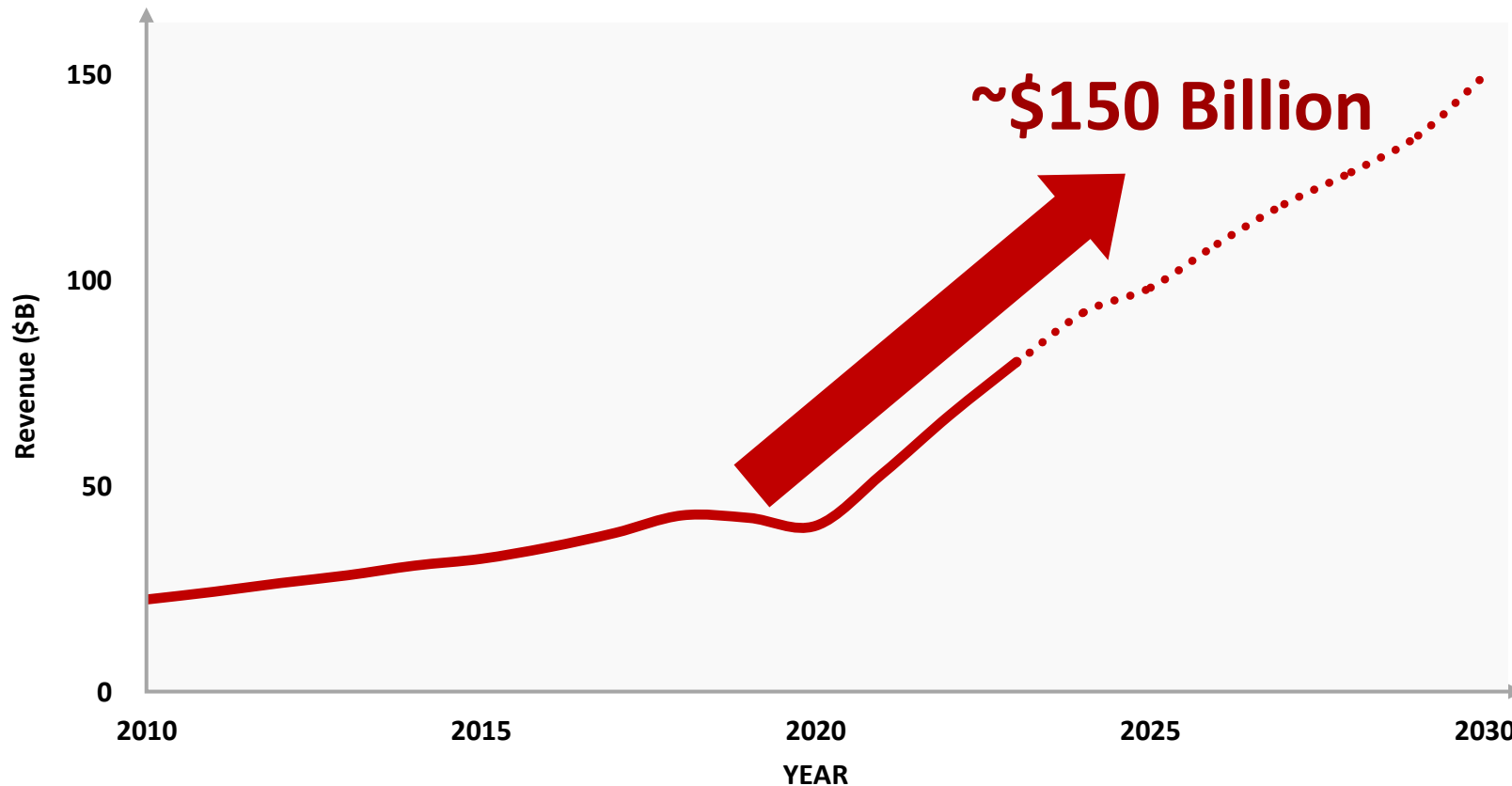
	Dimensity 9200	Dimensity 9300	Dimensity 9400
Process	N4	N4P	N3E
CPU	1x Cortex-X3 @ 3.05GHz 3x Cortex-A715 @ 2.85GHz 4x Cortex-A510 @ 1.8GHz	1x Cortex-X4 @ 3.25GHz 3x Cortex-X4 @ 2.85GHz 4x Cortex-A720 @ 2.0GHz	1x Cortex-X925 @ 3.63GHz 3x Cortex-X4 @ 3.3GHz 4x Cortex-A725 @ 2.4GHz
GPU	Immortalis G715	Immortalis G720	Immortalis G925
NPU	NPU690	NPU790	NPU890
Modem	4CC 300MHz R16, 7Gbps DL 3.5Gbps UL	4CC 300MHz R16, 7Gbps DL 3.5Gbps UL	4CC 300MHz R17, 10Gbps DL 3.5Gbps UL
GeekBench 6 (Single / Multi-Core)	1,949 / 5,281	2,239 / 7,538 (1.15X/1.43X)	2,874 / 8,969 (1.28X/1.19X)
GPU Perf. (TFlops)	3.6	4.0 (1.11X)	7.4 (1.85X)
GPU Perf. (Aztec)	67fps	92fps (1.37X)	115fps (1.25X)
AI NPU Perf. (TOPS)	30	35 (1.17X)	50 (1.43X)
Launch Date	Q4 2022	Q4 2023	Q4 2024

Outline

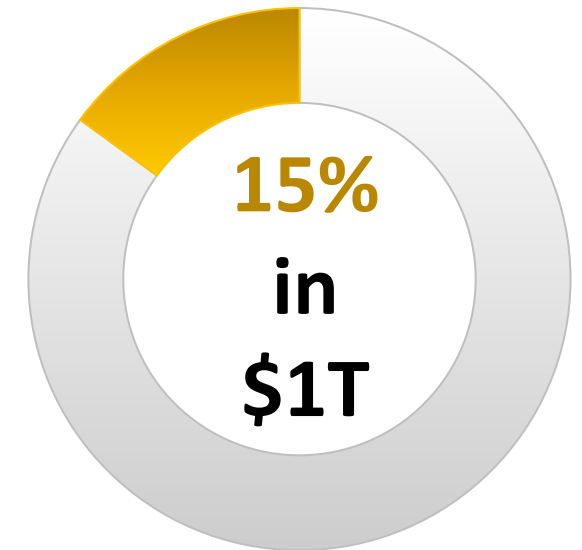
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AI Adoption Accelerates Automotive Silicon TAM

Worldwide Automotive Semiconductor Revenue



Automotive



Auto Megatrends: Safer, Smarter, Greener

~4.3X

ADAS / AD Level 2+ to 5
by 2030

~8.6X

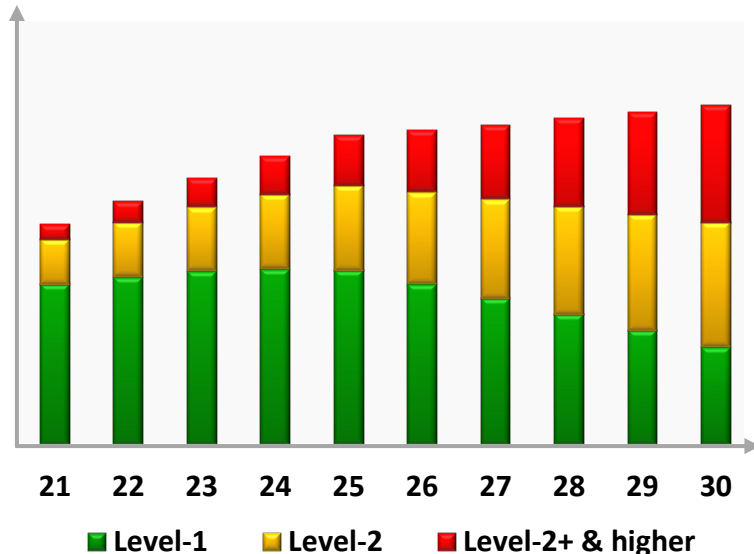
5G Installed in Vehicles
by 2030

~2.5X

Electric Vehicles growth
by 2030

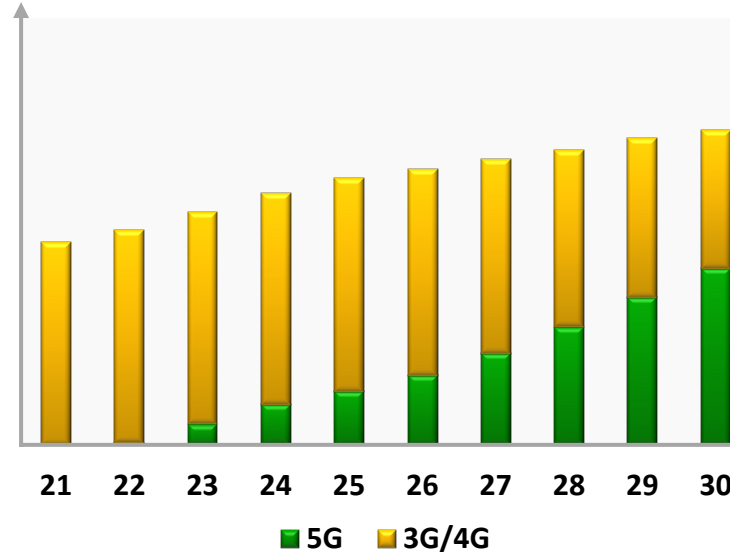
ADAS / AD

2030 Adoption Rate ~90%



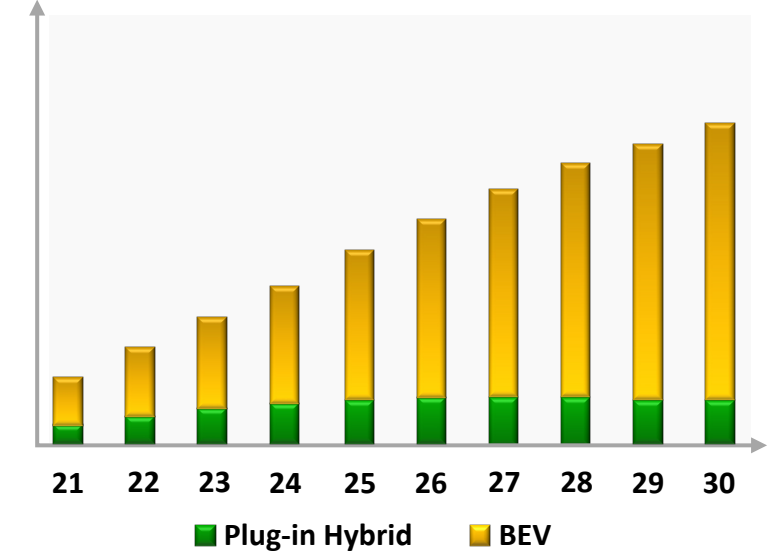
Connected & Digitization

2030 Adoption Rate >75%



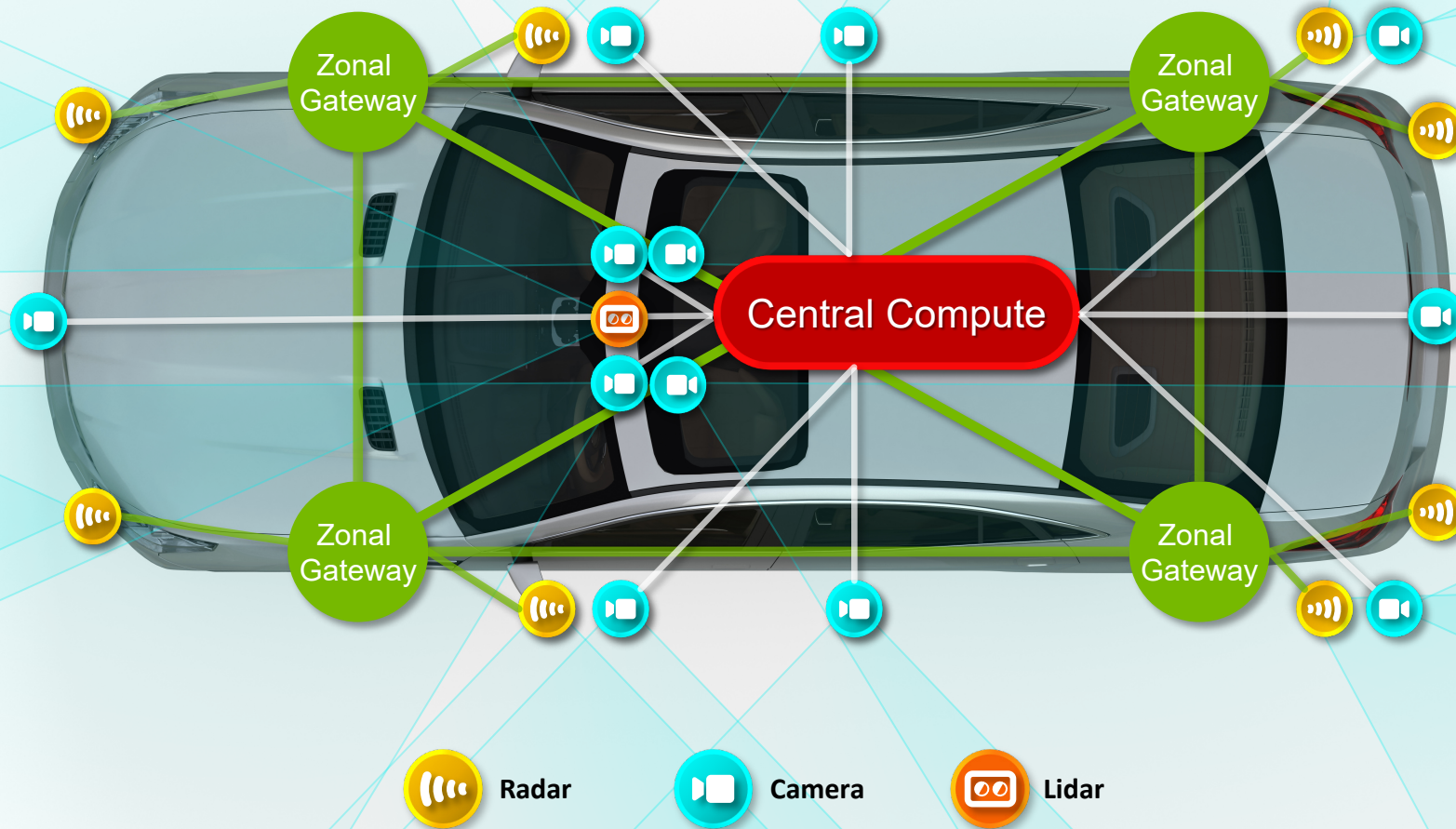
Electrification

2030 Adoption Rate >40%



New E/E Platform

Software Defined Vehicle with Zonal Architecture



AI Transforming Automotive Experience

Next-Gen AD & In-Car Experience Powered by AI



Robotaxi Service

Chatbot Assistant

City Pilot



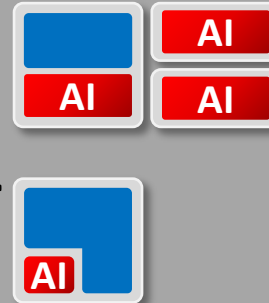
ADAS Domain
Controller



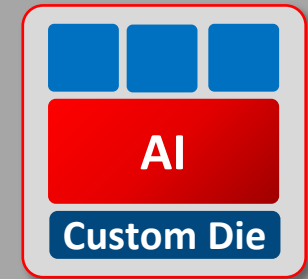
Cockpit Domain
Controller



Central
Computer



Auto
Chiplet



Evolution of Infotainment Drives Feature Integration

- Increasing compute power and graphic rendering capability motivate technology migration



Audio

+ BT
Audio

+ Video
RVC

+ Phone
Projection

+ Around
View
Monitoring

+ Digital
instrument
Cluster, HUD

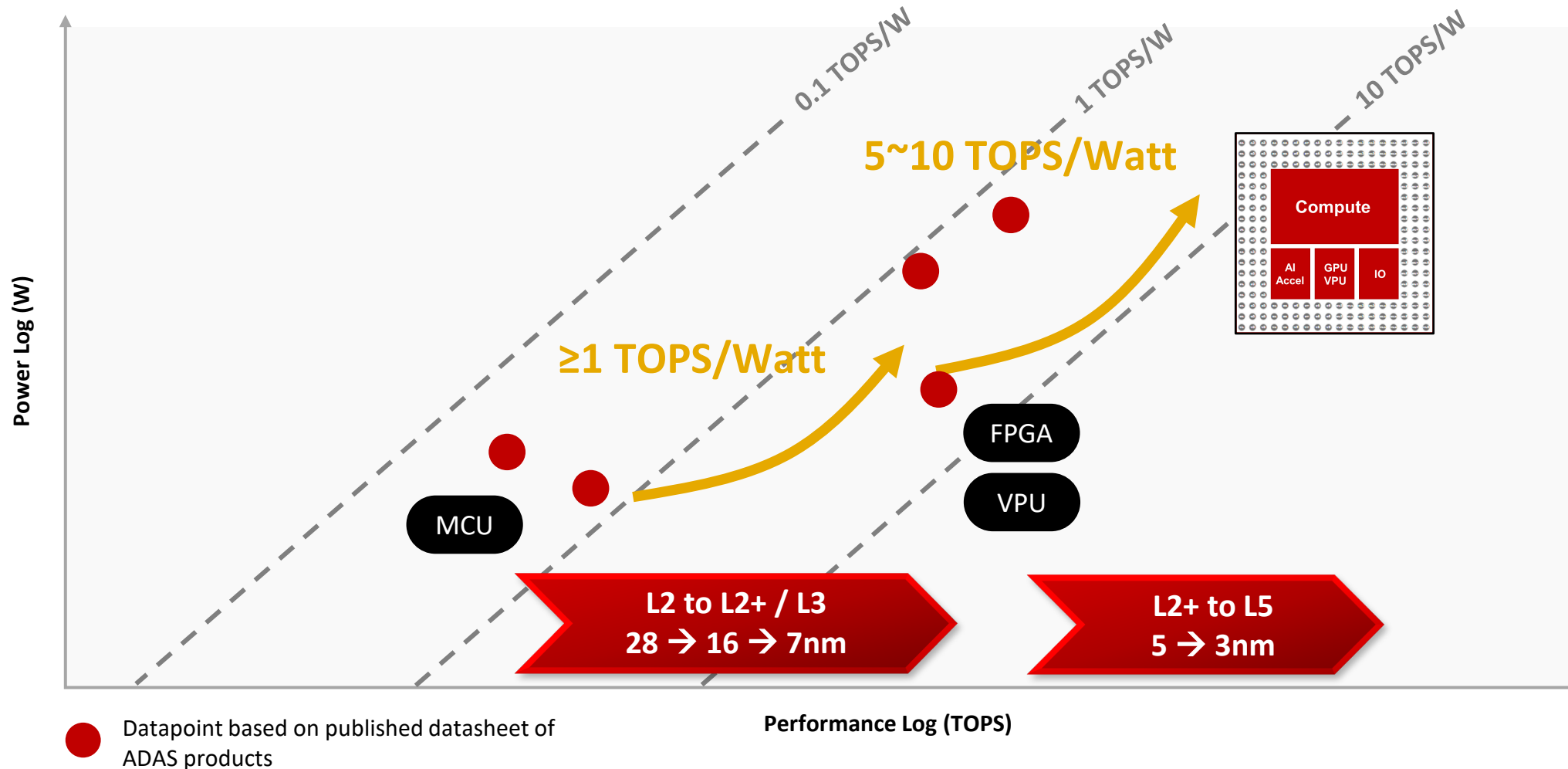
+ in cabinet
monitoring

+ Multi
display

+ Safety
Features

IVI Cockpit	Display	0	0	1	1	1	2-3	2-3	3-9	3-9			
	CPU KDMIPS	-	<1	<5	5-10	10-50	50-100	50-100	100-200	>200			
	GPU GFLOPS	-	-	<30	30-100	100-500	500-3K	500-3K	500-3K	>3K			
	NPU TOPS	-	-	<1	<5	<5	5-30	5-30	5-30	>30			
		40nm		→	28nm		→	16nm		→	7nm	→	3nm

Advancing ADAS Requires Energy-efficient Compute



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AI Infusion is Future of IoT

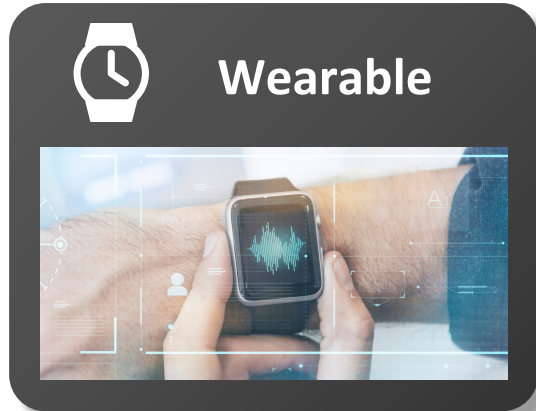
AIoT Market CAGR >10%



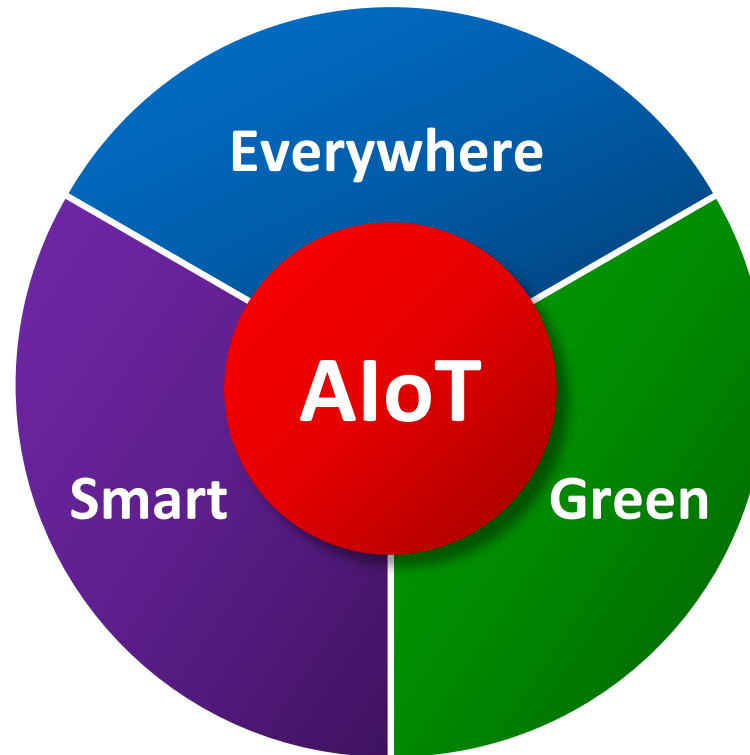
AIoT Making the World a Better Place

Edge AI Brings More Convenient, Healthier and Greener Life

Pervasive Connectivity
Links **Everywhere**



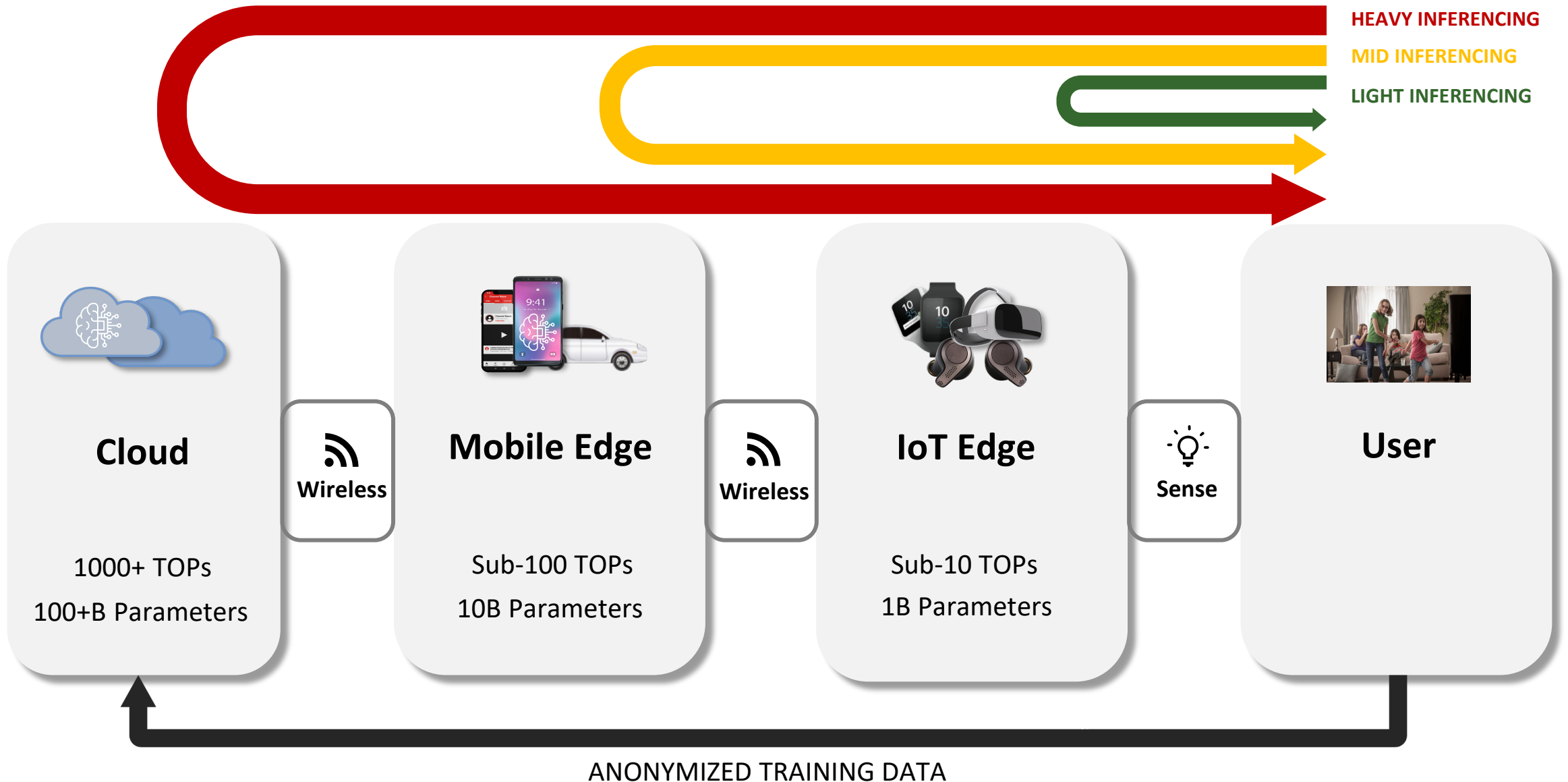
Advanced Tech
for **Smart** AI & ML



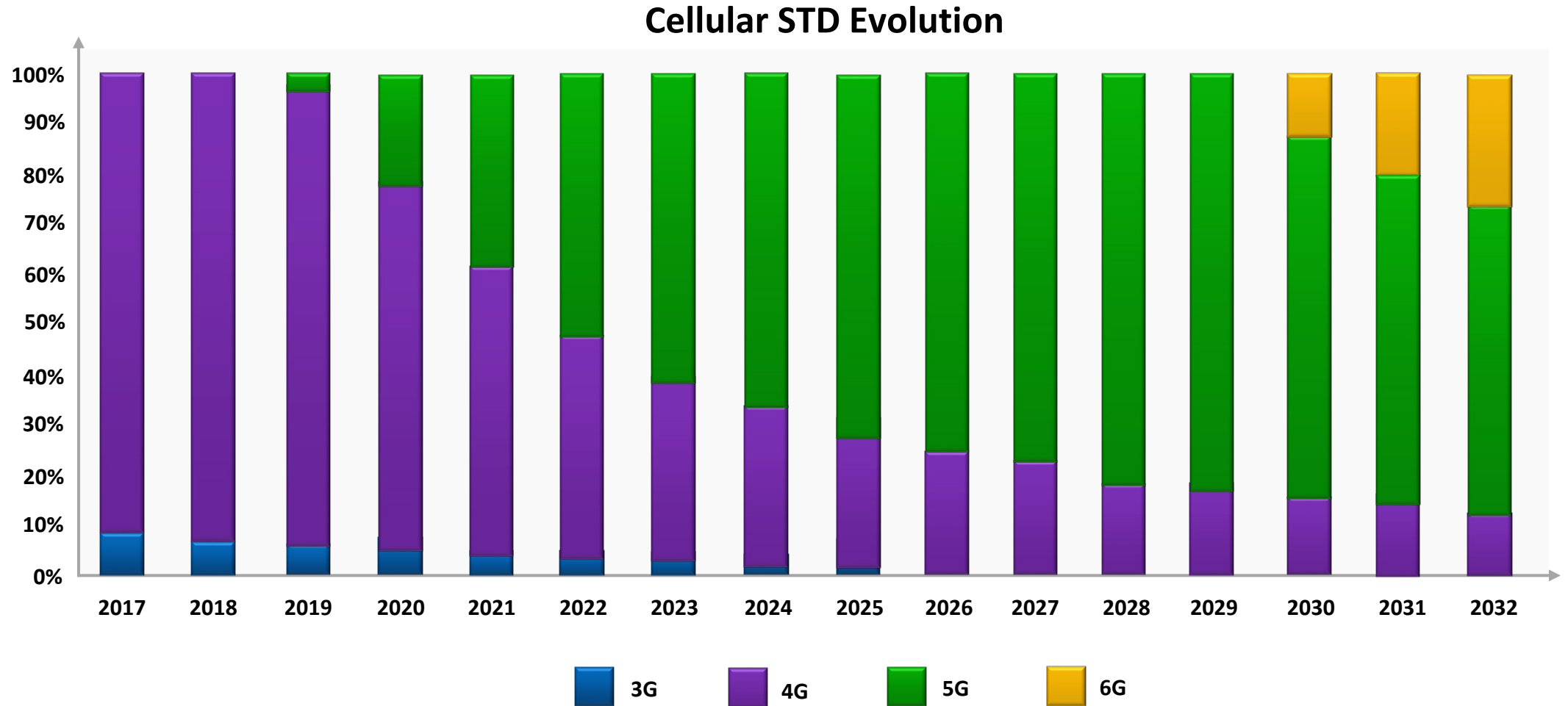
Ultra Low Power
for **Green** IoT



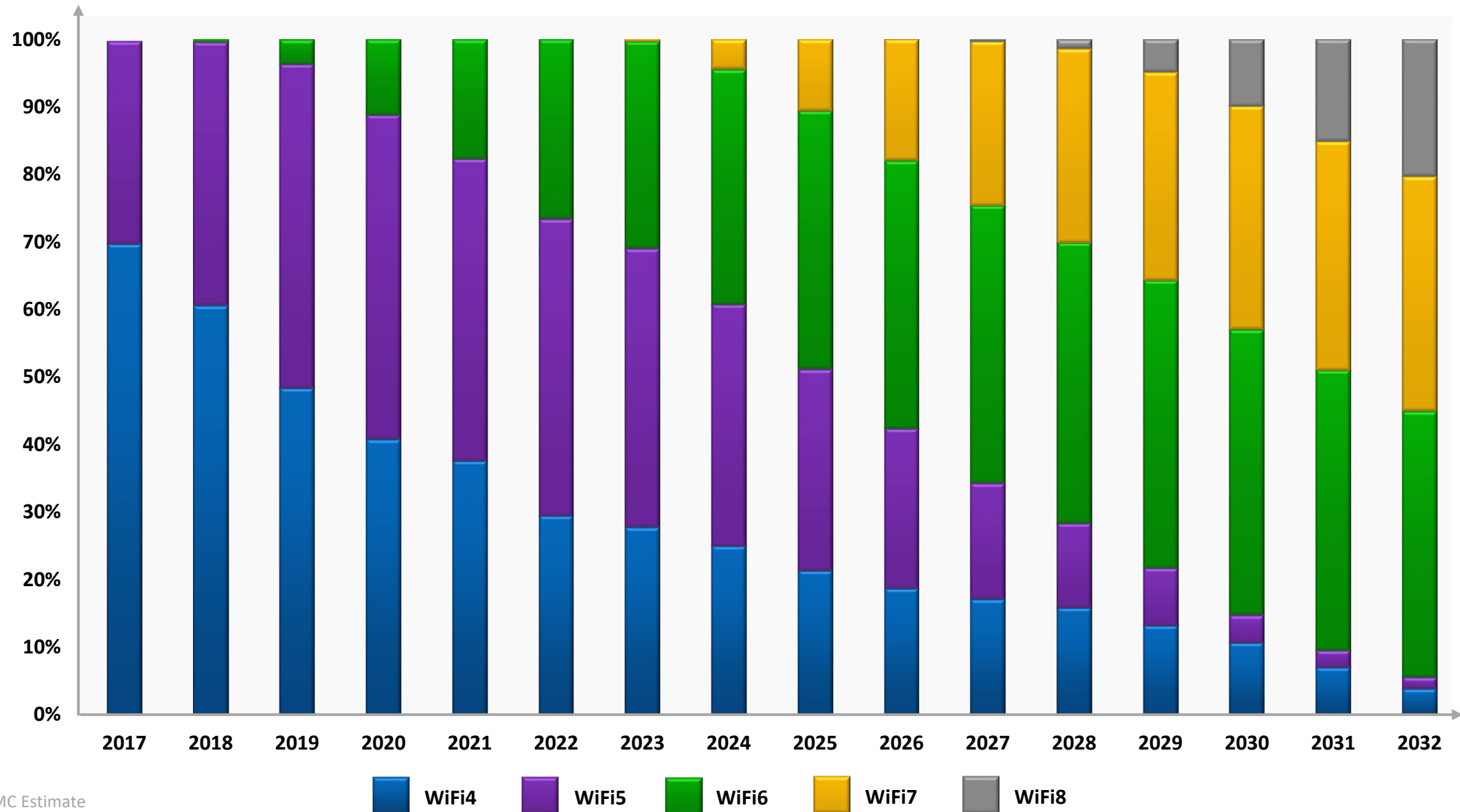
Wireless Connectivity Essentiality to AI Experience



Cellular Connectivity Evolves Toward 6G

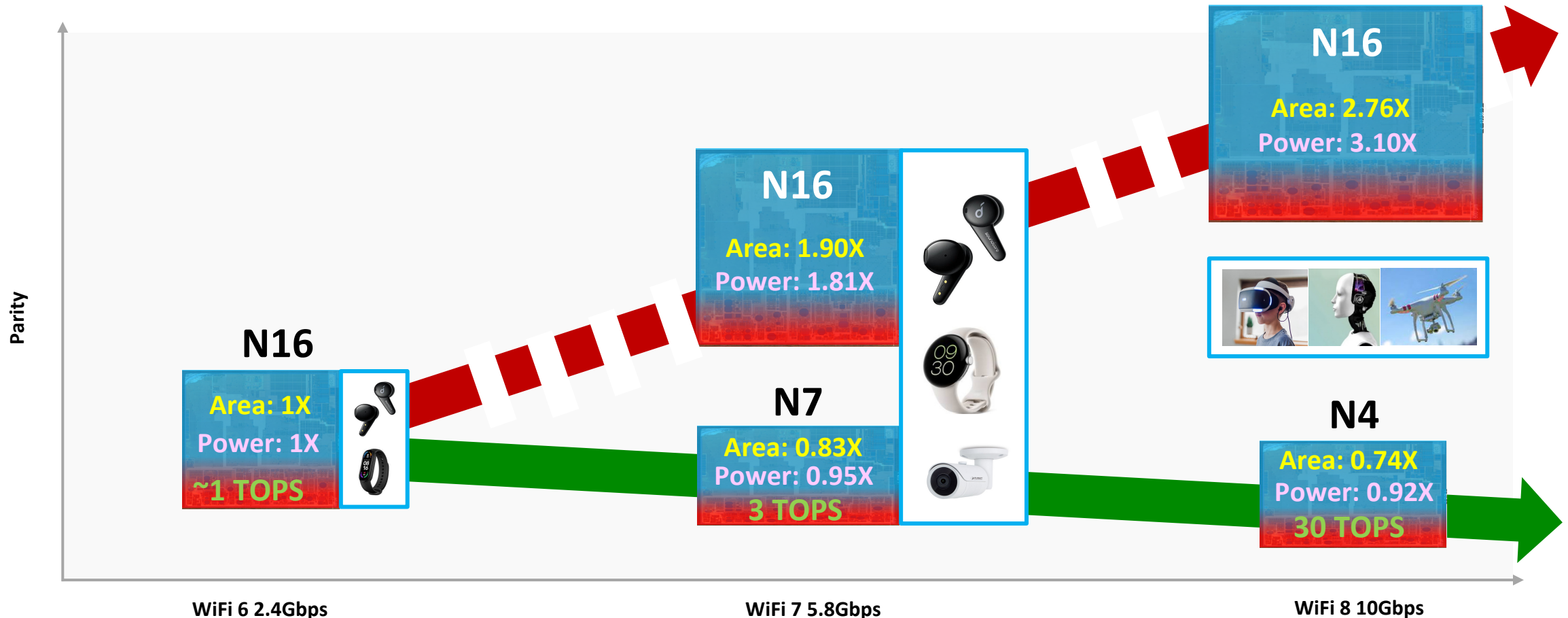


Continuous WiFi Standard Evolution



WiFi Evolution Drives RF Technology Migration

- WiFi evolution increases data rate and chip area & power
- Technology migration addresses form factor and battery life through generations



Product Requirement for AI Era

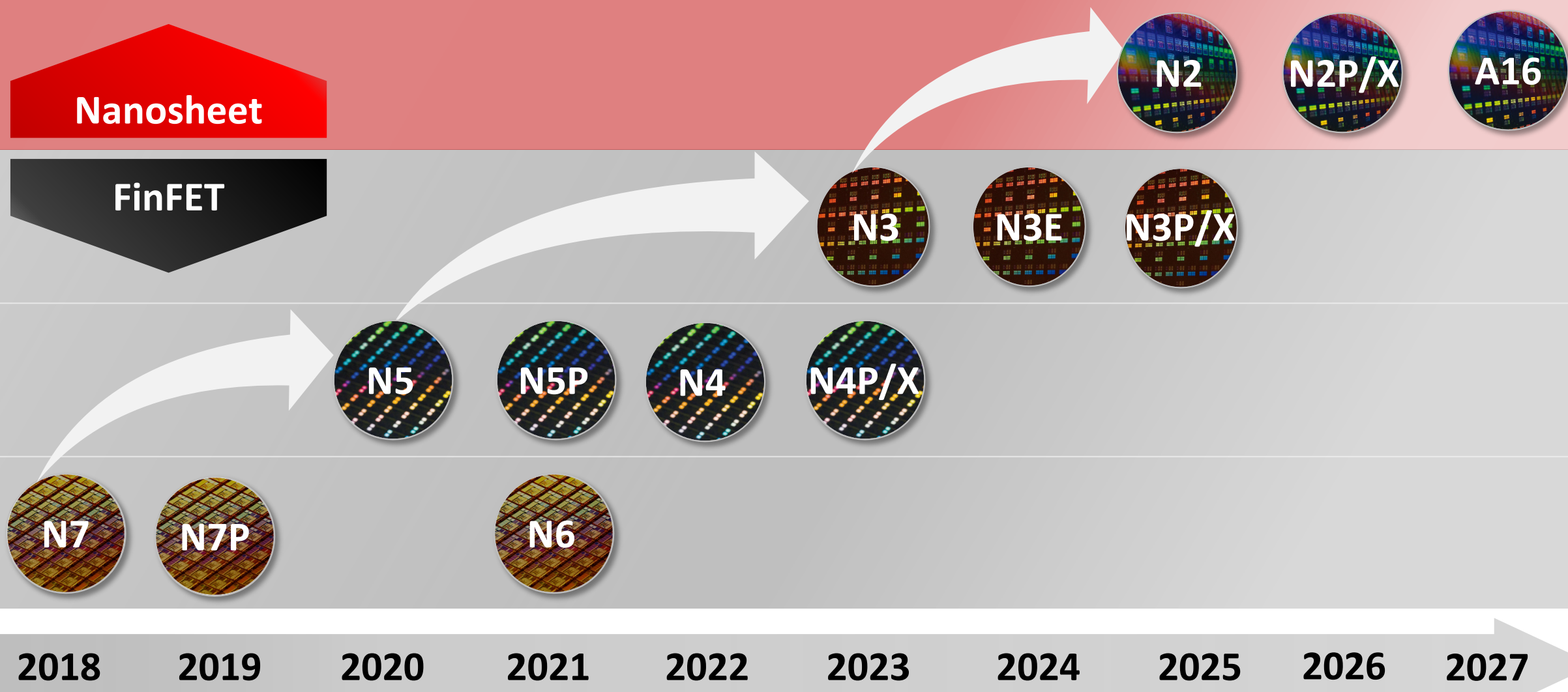
- **Energy efficiency becomes THE most important performance index for a wide range of applications**
- **Maximize energy efficiency to maximize compute under power constraint**
 - **Drive energy efficiency in compute, data movement and power delivery**
- **Minimize distance of data movement, increase memory bandwidth, minimize energy per bit and reduce form factor to save power**
 - **Heterogeneous integration of logic, memory, photonics, power delivery/voltage regulation**
- **Advance thermal solution for effective heat removal**

Outline

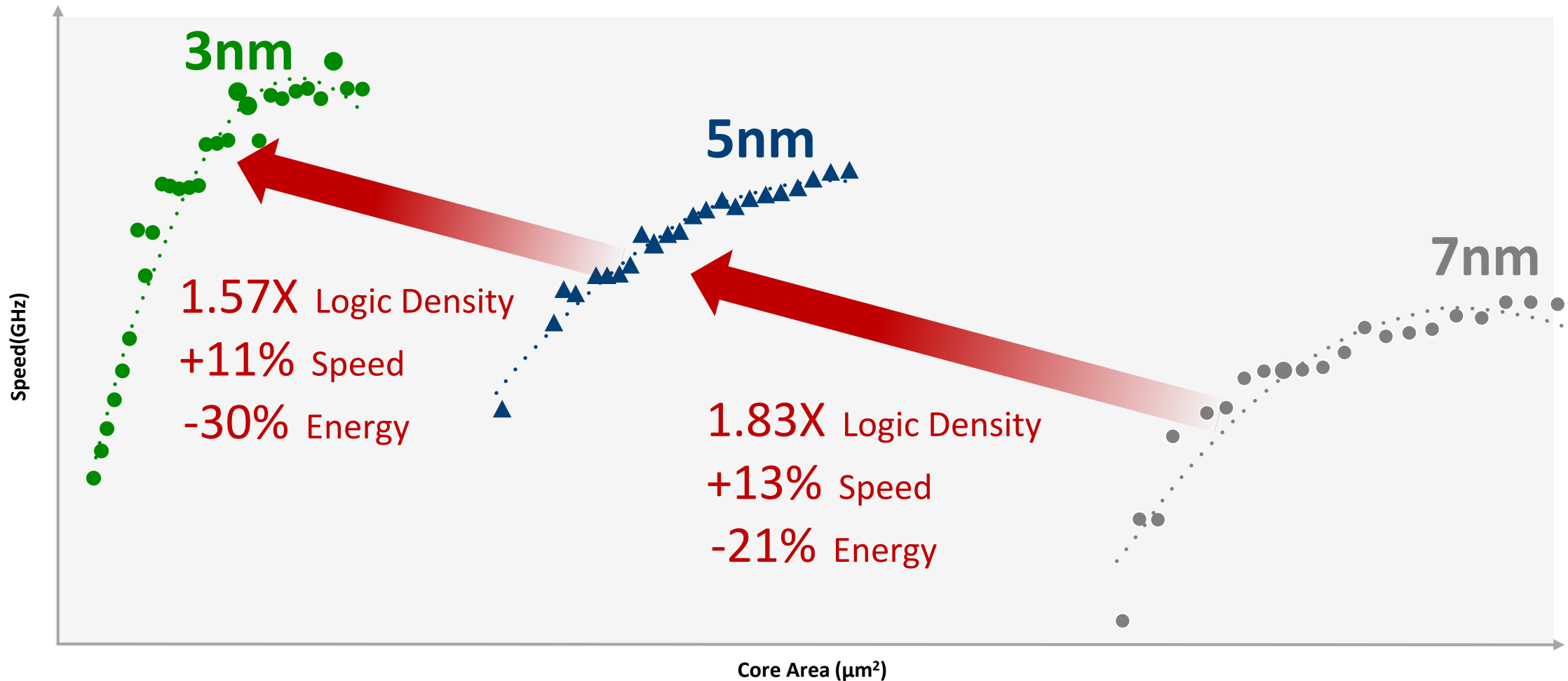
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Advancement in Logic Technology

- Logic technology continues to advance at steady pace for improved density and energy efficiency
- Technology enhancement in Year 2 and beyond enables better products with incremental investment

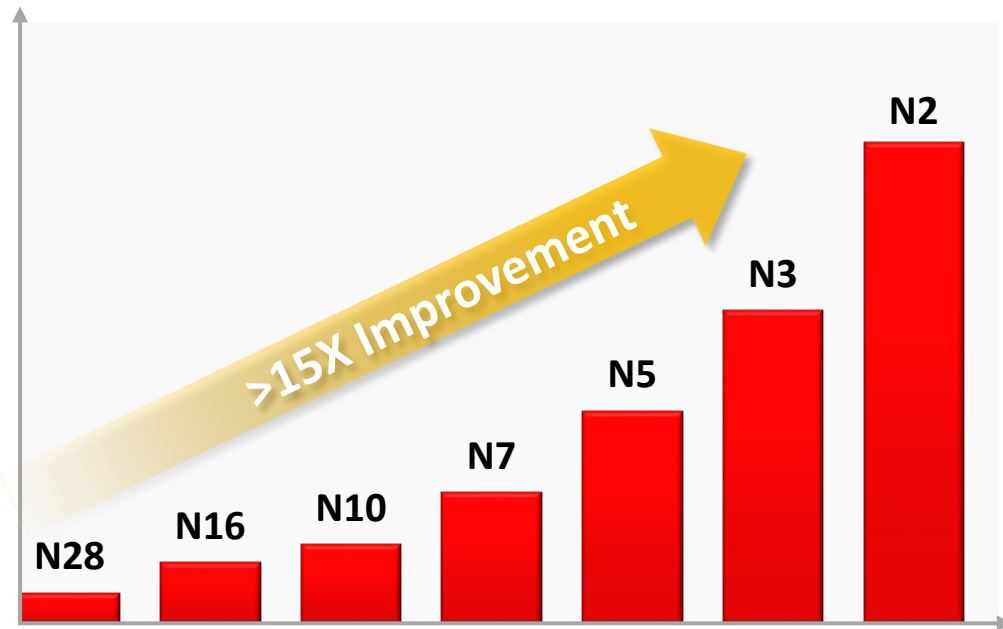


Technology Advancement Drives PPA Scaling

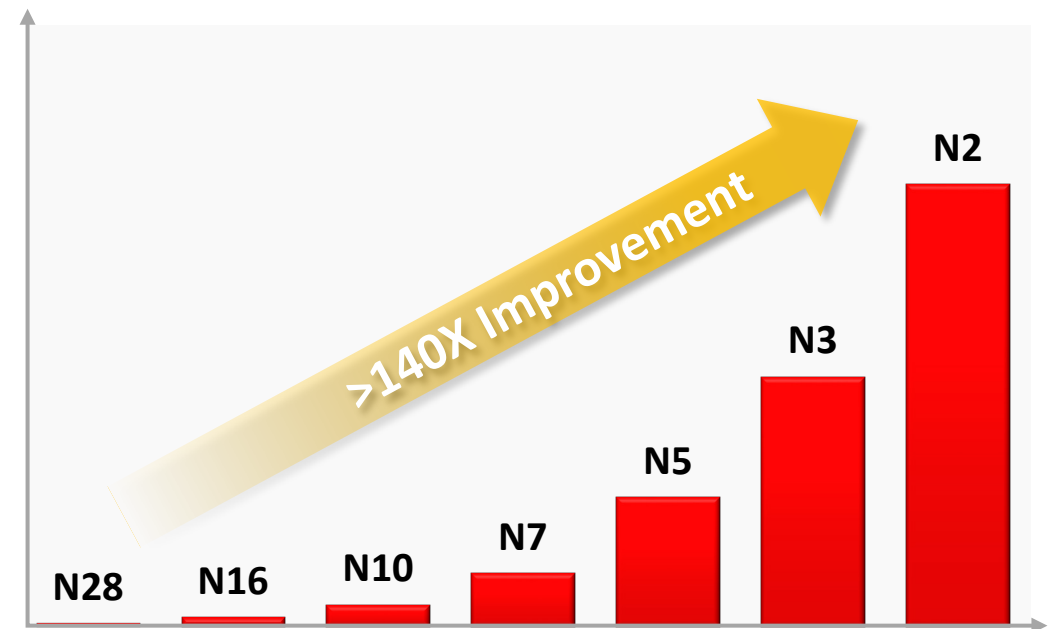


Technology Advancement Enables Energy Efficient Compute

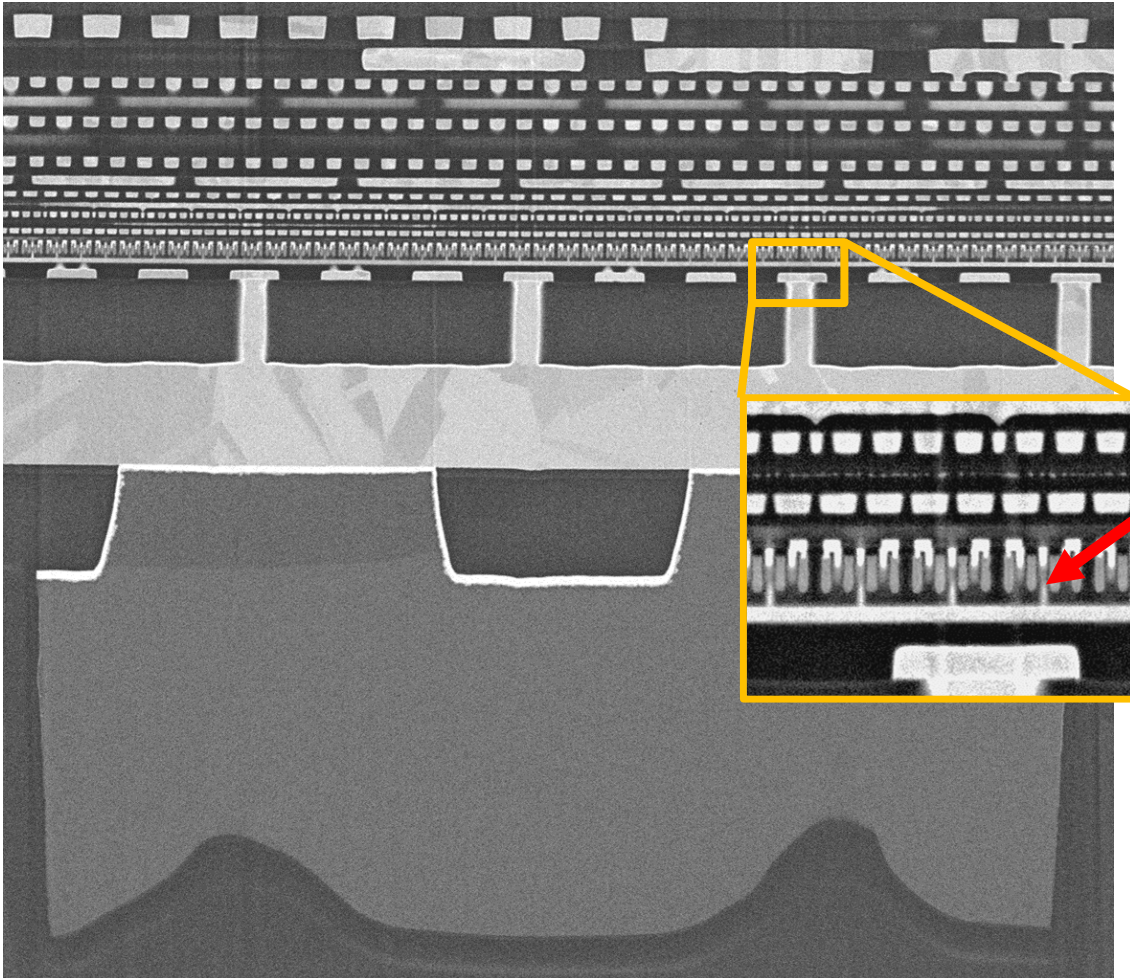
Power Efficiency



Perf / Watt / mm2



State-of-the-Art Nanosheet-based 2nm Logic Technology



BEOL process and materials offer up to 10% R*C delay reduction and friendly design rules to enable 3~4% logic density gain

Nanosheet devices offer excellent power efficiency from structural and DTCO innovations

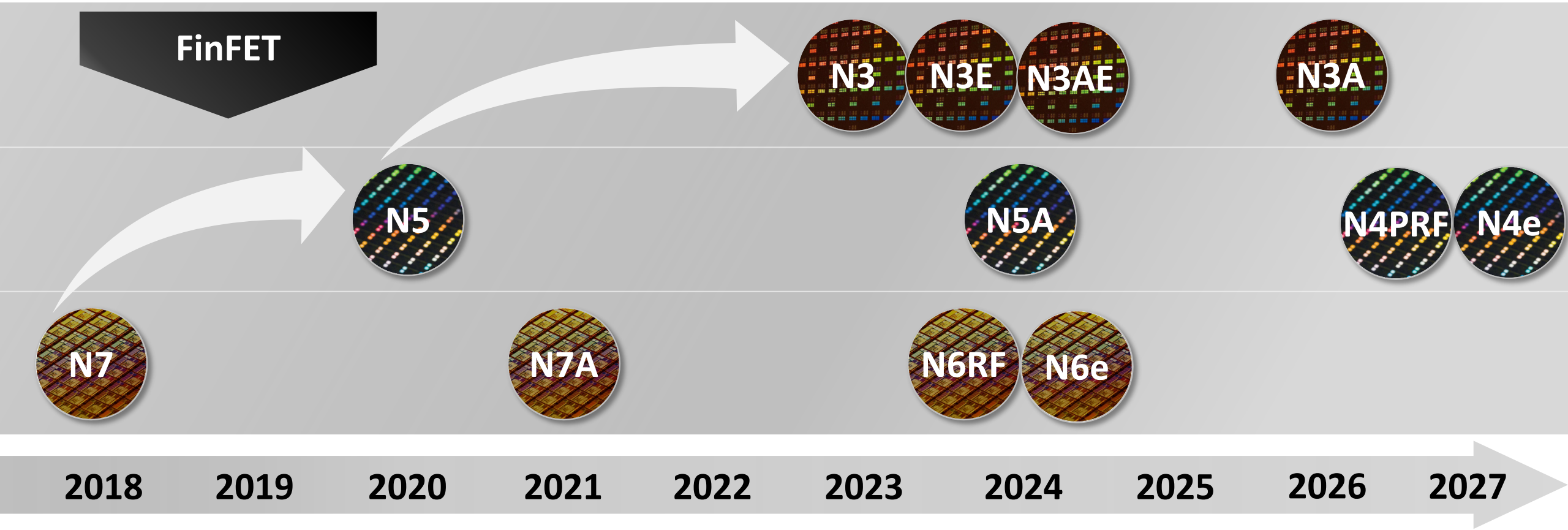
Backside Contact preserves gate density and device width flexibility

Backside Metallization

- Improves power delivery and reduce IR drop
- Improves chip density and performance by dedicating front-side routing to signals

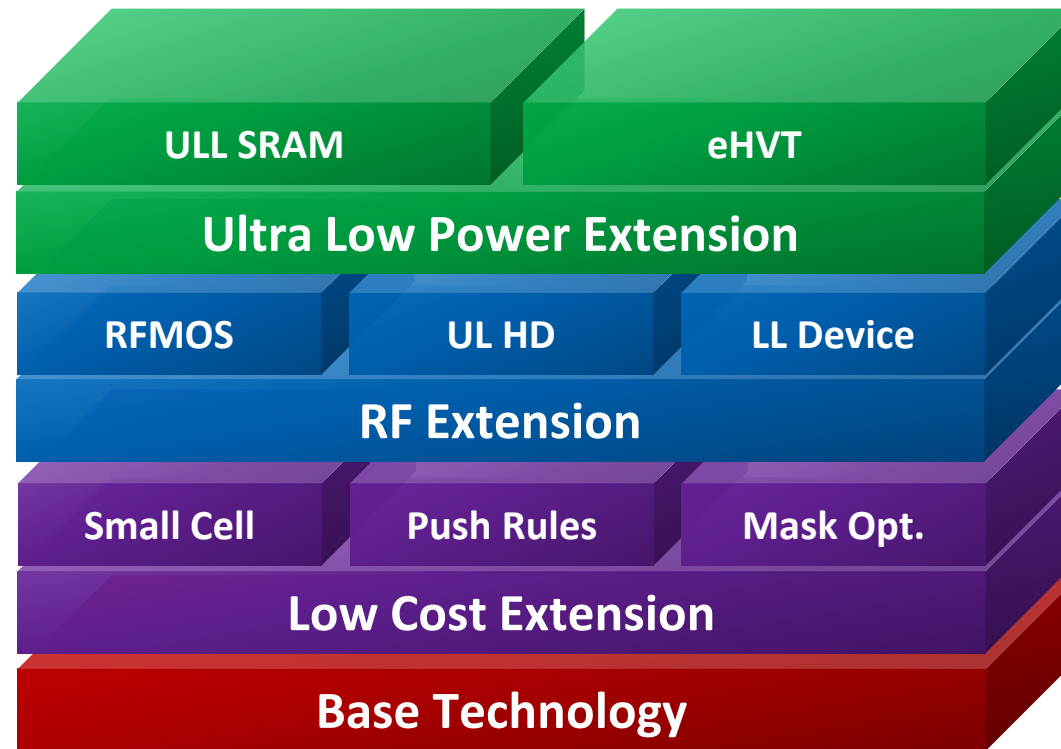
Application Specific Technology Portfolio

- Extensions from baseline technology to meet diverse application requirements



One Platform To Maximize Customer Investment

- Application-specific technology features compatible with baseline are added over time
- One Platform ensures reuse of existing IPs and sharing of new IPs across products

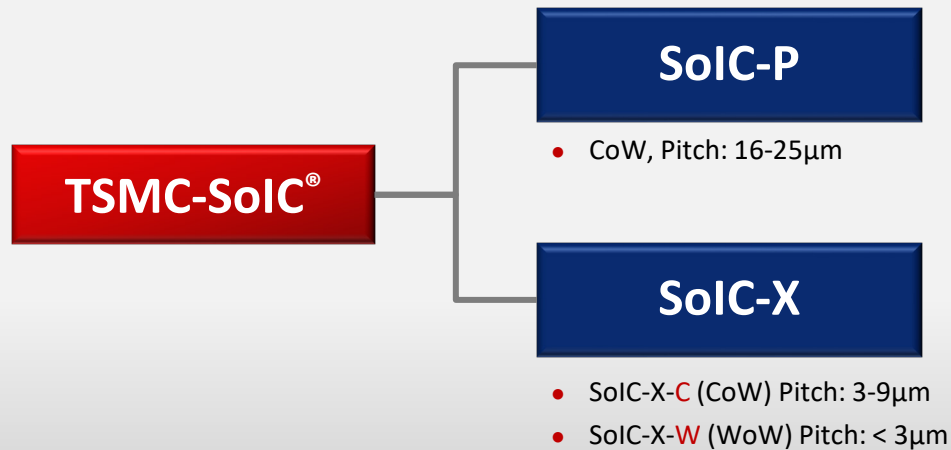


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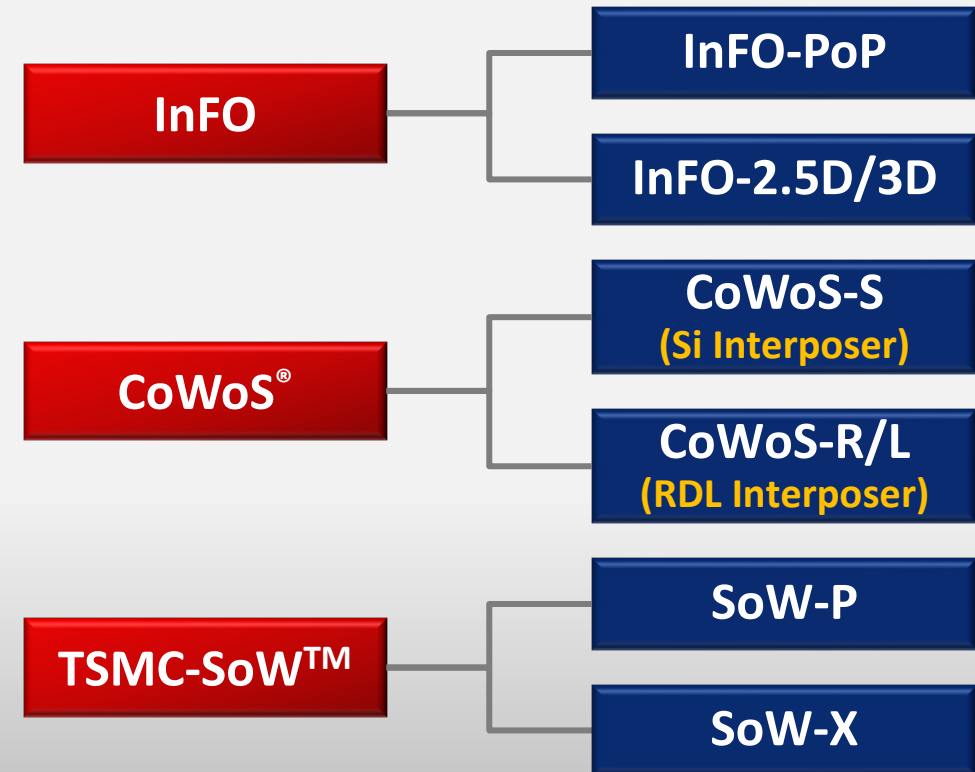
System Integration Technologies

3D Si Stacking



SolC: System on Integrated Chips

Advanced Packaging



CoWoS: Chip on Wafer on Substrate

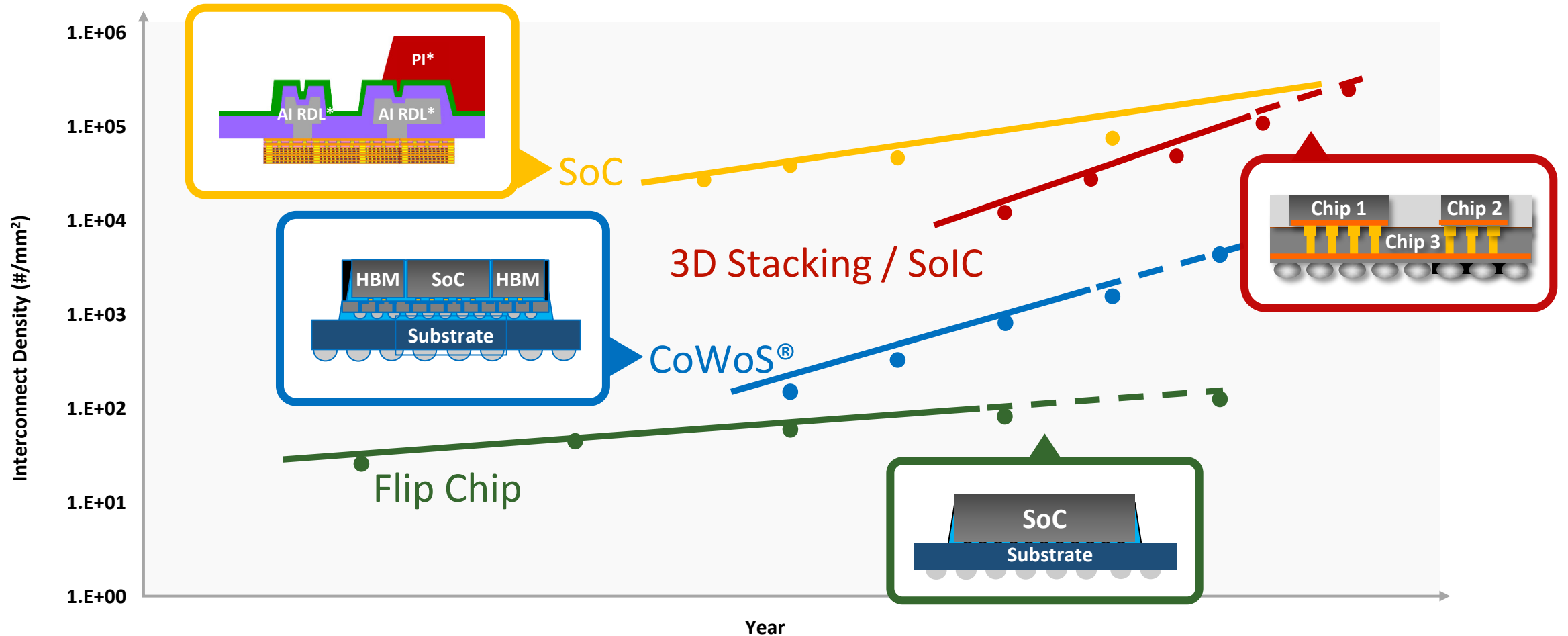
InFO: Integrated Fan-Out

PoP: Package on Package;

RDL: Redistribution Layer

SoW: System on Wafer

Advanced Packaging Technologies Enable Interconnect Density Scaling

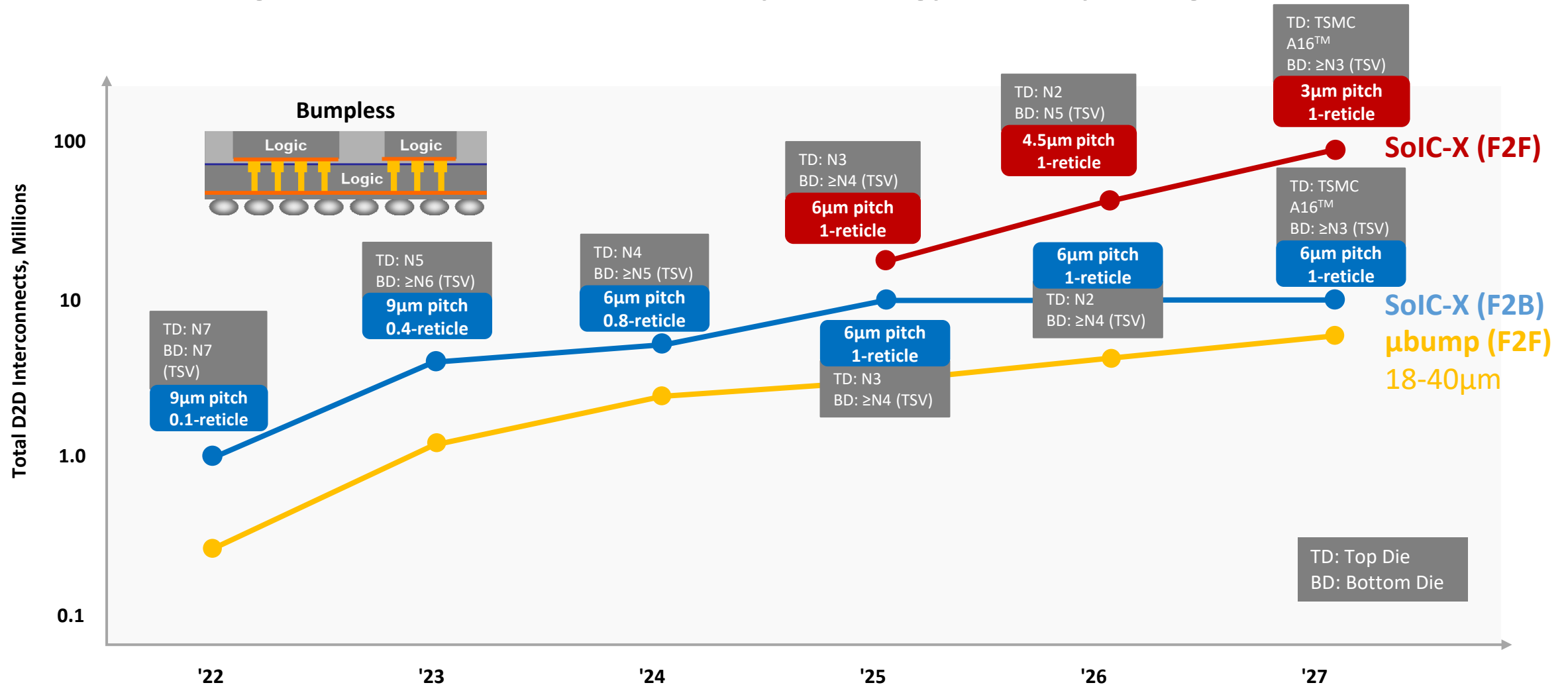


Source: TSMC

*: PI: Polyimide; AI RDL: Aluminum Redistribution Layer

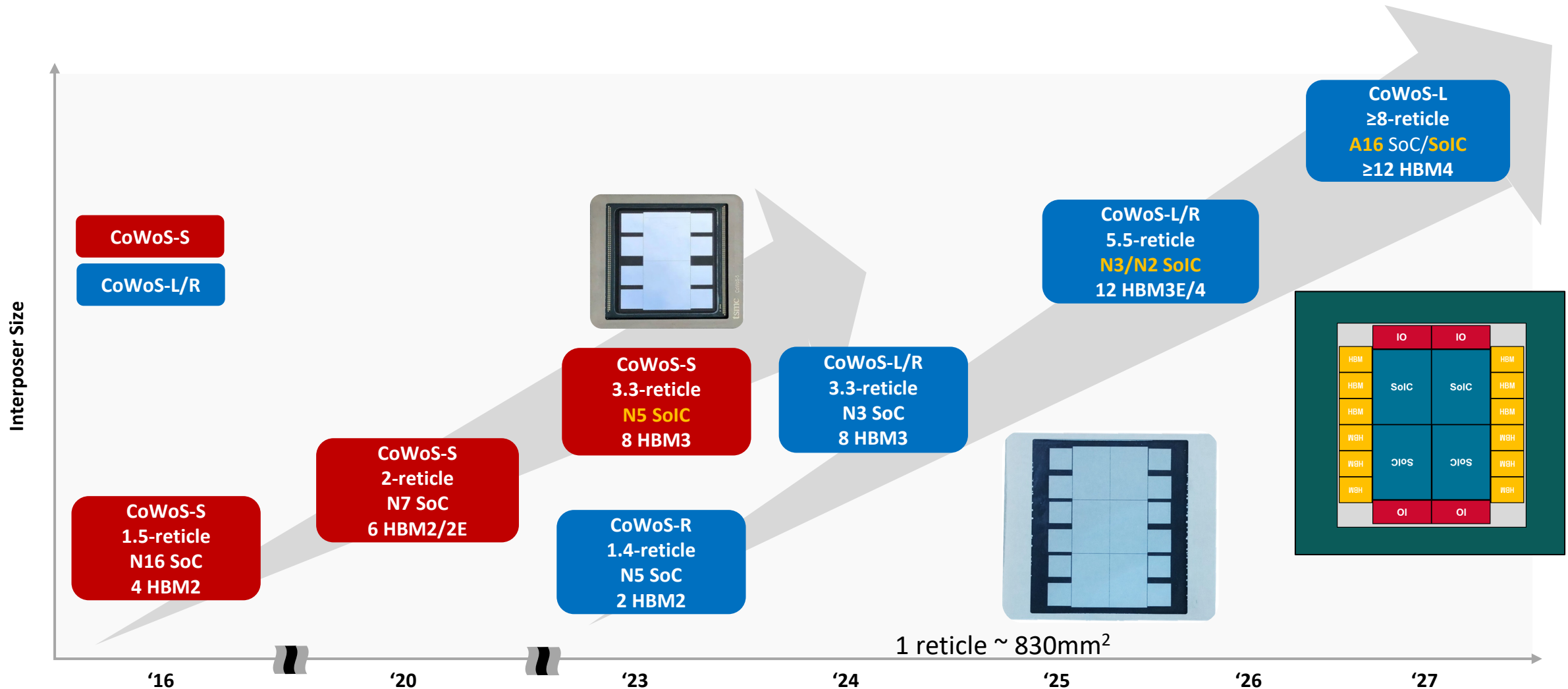
3D Stacking Technology Progression

- First 3D-stacked datacenter CPU shipped in '22 and AI GPU in '23
- 3D stacking offers a new dimension of density and energy efficiency scaling



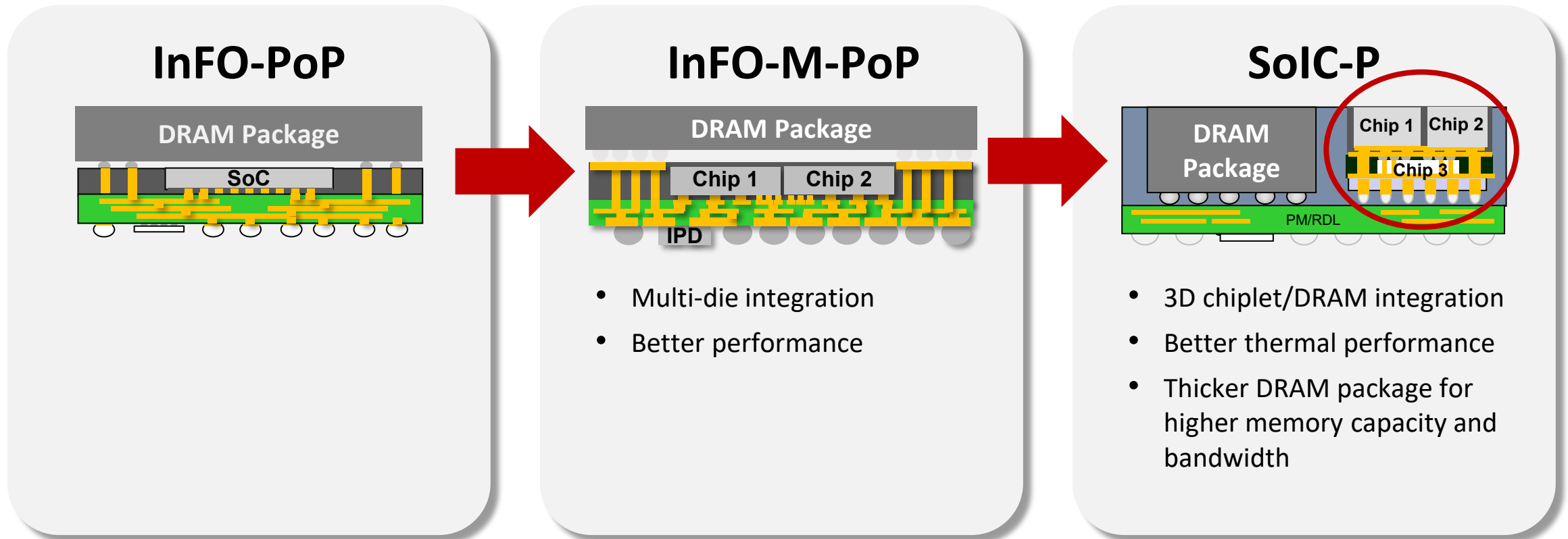
2.5D Integration Enables Next-Gen AI Compute

- 2.5D technology envelope is rapidly expanding to address integration needs of future AI compute



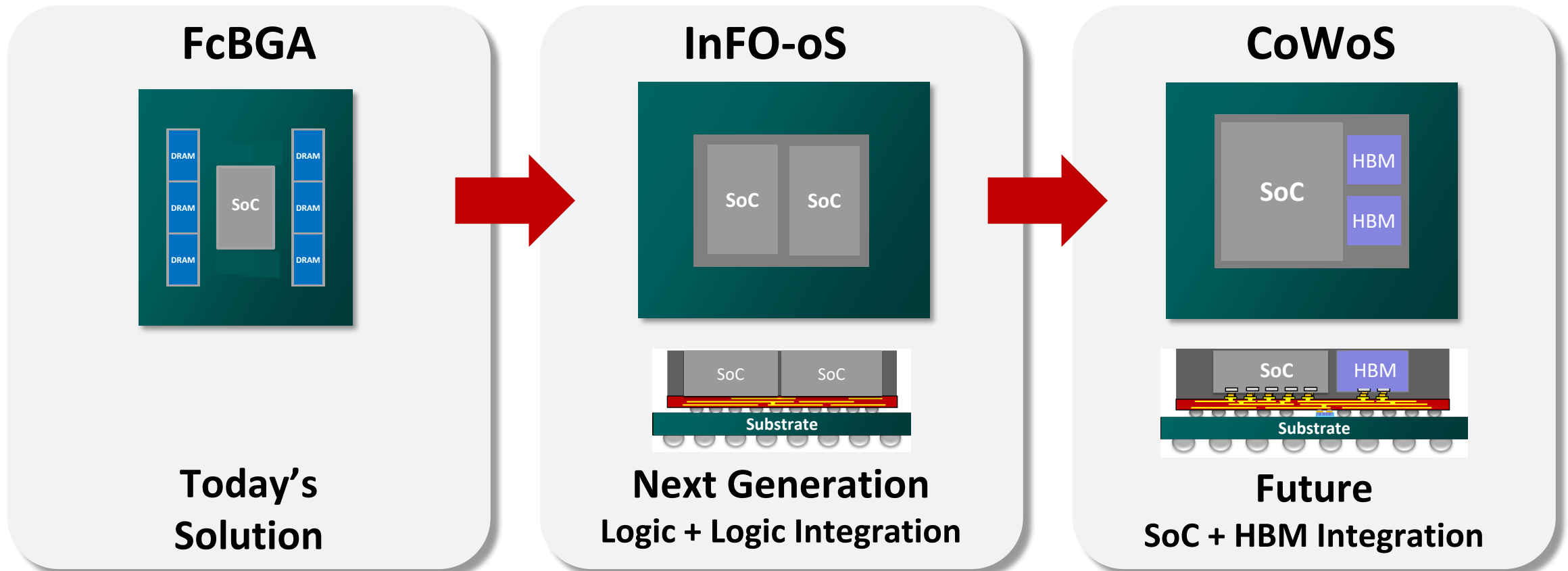
Advanced Packaging Trend for Mobile AI

- InFO-M-PoP enables chiplet architecture through heterogeneous integration
- SoIC-P enables 3D chiplet architecture for enhanced performance and keeps form factor under check



Advanced Packaging for Automotive AI

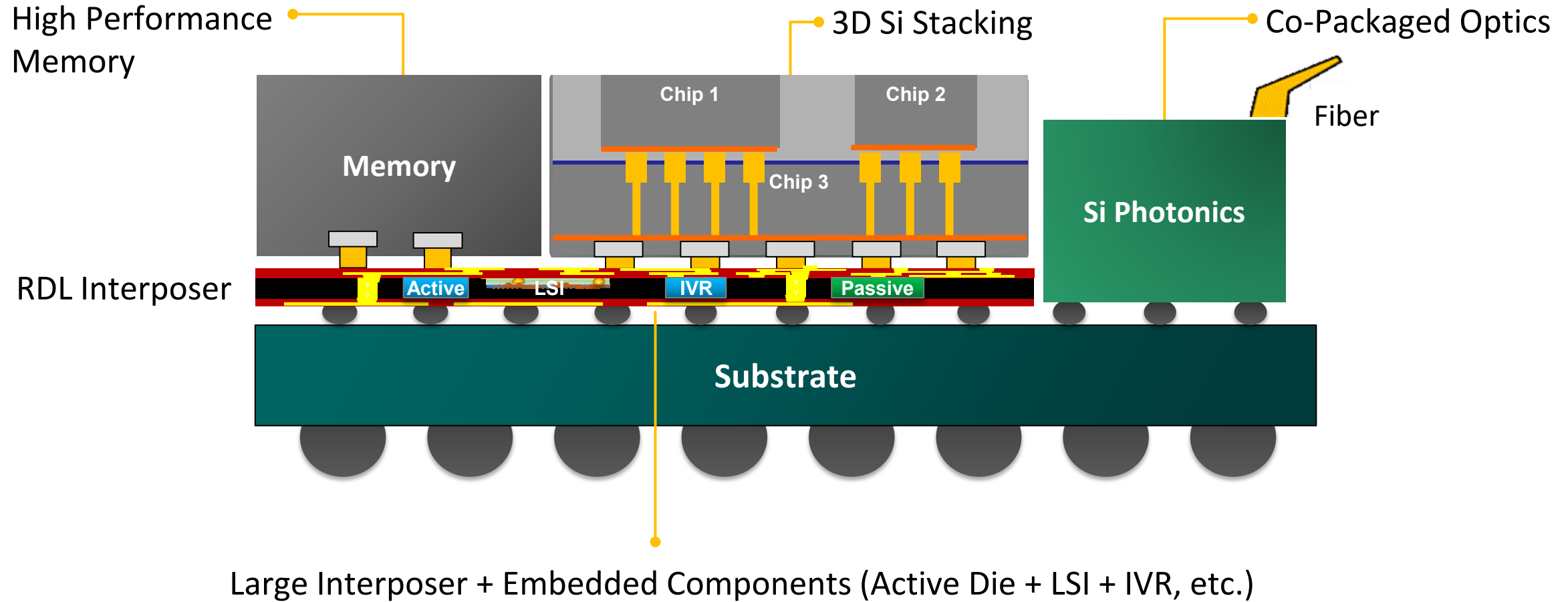
- Chiplet integration boosts ADAS performance and provides design flexibility
- InFO-oS for logic + logic and CoWoS for logic + HBM integration



Outline

- Long-term market overview
- AI megatrend and impact on
 - Data Centers
 - Edge devices
 - Automotive
 - IoT
- Advanced Logic
- Advanced Packaging
- **System-Technology Co-Optimization (STCO)**
- Summary

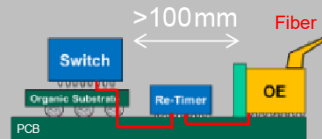
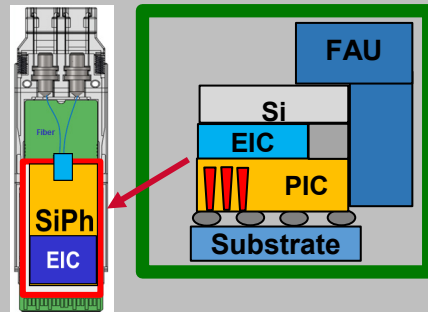
Enabling Tomorrow's AI with Technology Platform



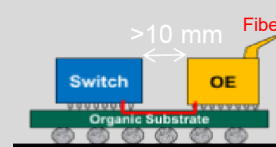
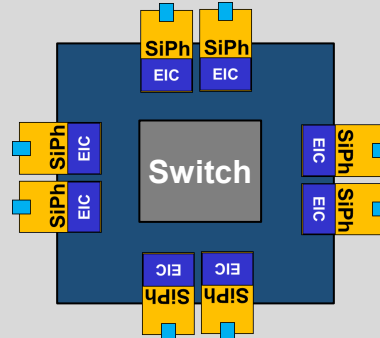
Silicon Photonics Revamps Data Transmission

- Vertically stacked Optical Engine (OE) enables high packing density and low system power
- Board, package and interposer level OE integration offers progressive form factor and power benefit

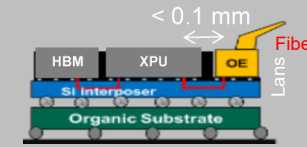
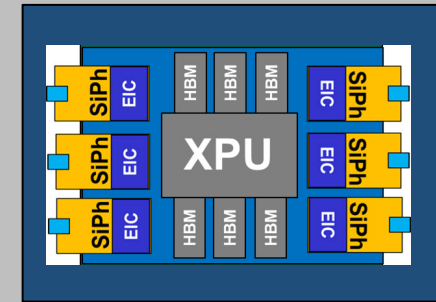
OE on PCB (Pluggable)



OE on Substrate



OE on Interposer



Power Efficiency
Latency

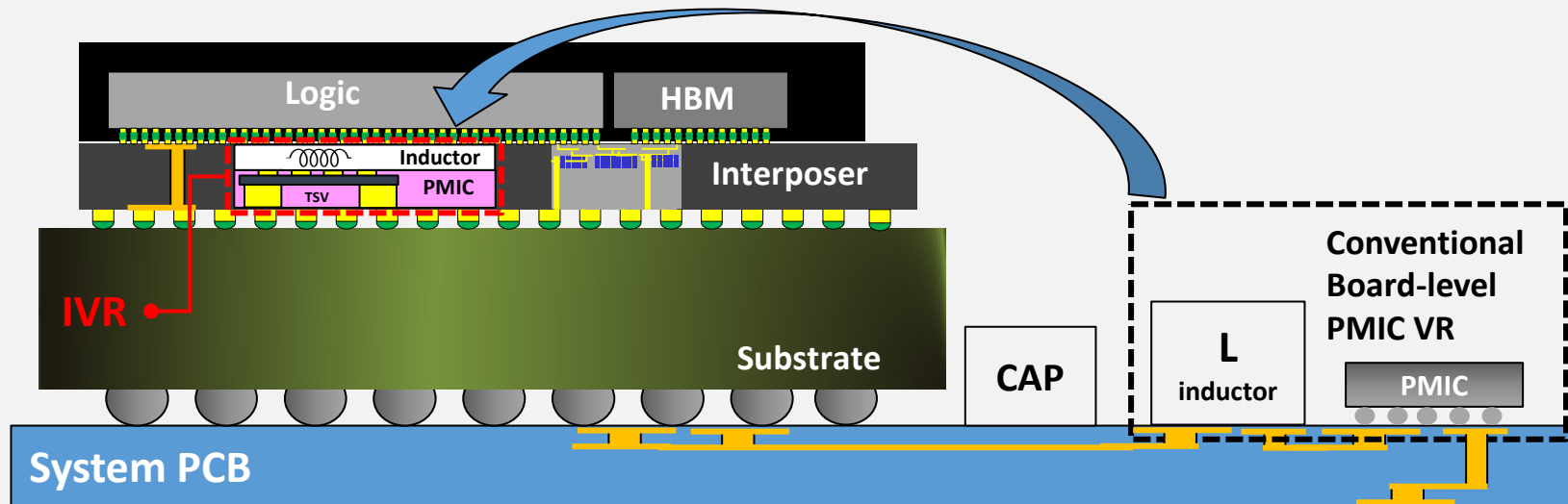
> 10pJ/bit
1X

> 5pJ/bit
< 0.1X

< 1pJ/bit
< 0.05X

Next-Gen Power Delivery: Integrated Voltage Regulator

- AI is compute-intensive and needs advanced power delivery solution
- Voltage regulator needs to supply high power density with ultra-fast response and low loss in a compact form factor
- FinFET + Integrated Inductor + 2.5D Integration enable high-performance voltage regulator



Summary

- **Propelled by AI, semiconductor industry will reach \$1T by 2030**
- **AI has transformative impact on datacenter, edge device, automotive and IoT products**
- **Energy efficiency becomes the most important performance index for AI-infused platforms moving forward**
- **Advancement in logic, memory and packaging technology plots a path of continuous density and energy efficiency scaling to meet AI-driven requirements**
- **STCO holistically enables tomorrow's AI with integrated technology platform**